

Link Between the Differential Cryptanalysis and Linear Approximations over Finite Abelian Groups And Its Applications

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Abstract. In recent years, progress in practical applications of multi-party computation (MPC), fully homomorphic encryption (FHE), and zero-knowledge proofs (ZKP) motivates people to explore symmetric-key cryptographic algorithms, as well as corresponding cryptanalysis techniques (such as differential cryptanalysis, linear cryptanalysis), over finite Abelian groups or prime fields $\mathbb{F}_p$ for large p . In this paper, we establish the links between linear cryptanalysis and differential cryptanalysis over general finite Abelian groups. As the first application, we revisit linear cryptanalysis and give general results of linear approximations over arbitrary finite Abelian groups. More precisely, we consider the *linearity*, which is the maximal non-trivial linear approximation, to characterize the resistance of a function against linear cryptanalysis. This thereby generalizes the work of Pott in 2004 and completes the generalization of Sidelnikov–Chabaud–Vaudenay’s bound from $\mathbb{F}_2^n$ to finite Abelian groups. As the second application, we give an exact expression for the correlation of differential-linear approximations over arbitrary finite Abelian groups ($\mathbb{F}_p^n$) under the sole assumption that the two parts of the cipher are independent of each other. In particular, we completely generalize the differential-linear cryptanalysis from $\mathbb{F}_2^n$ to arbitrary finite Abelian groups ($\mathbb{F}_p^n$).

Keywords: Linear Cryptanalysis, Differential Cryptanalysis, Finite Abelian Groups, Sidelnikov–Chabaud–Vaudenay’s Bound, Linear Approximations, Differential-linear Approximations.

1 Introduction

Differential cryptanalysis, linear cryptanalysis and differential-linear cryptanalysis are the three of common methods in cryptanalysis of symmetric-key ciphers.

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Differential cryptanalysis was proposed by Biham and Shamir [BS90] in 1990, and has become one of the common methods in cryptanalysis of symmetric-key ciphers that exploits the differences of input and output pairs to recover the secret key of symmetric-key ciphers. Linear cryptanalysis was proposed by Matsui [Mat93] in 1993. Later, differential-linear cryptanalysis was proposed by Langford and Hellman [LH94] in 1994 as a further development of the idea of linear approximation.

For linear cryptanalysis, the main idea is to find linear relations, known as linear approximations, between the parity bits of the plaintext, the ciphertext, and the secret key, with high probability. Since Matsui’s work, linear cryptanalysis has attracted intensive attention and has been developed to several extensions and variants, for example, see [JR94, BLNW12, HCN08, BCQ04, DP08, HCN09, BW12].

In order to characterize the resistance of symmetric-key primitives against linear cryptanalysis, several metrics are proposed. Nyberg [Nyb91] and Chabaud and Vaudenay [CV94] use the linearity (linear-resistance) λ_F of Boolean vector functions $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ as

$$\lambda_F = \max_{a \in \mathbb{F}_2^n; b \neq 0^m, b \in \mathbb{F}_2^m} 2^{-n} \left| \sum_{x \in \mathbb{F}_2^n} (-1)^{a \cdot x \oplus b \cdot F(x)} \right|,$$

where $\cdot$ denotes the inner product of the two vectors over $\mathbb{F}_2$. Thanks to the link between differential cryptanalysis and linear approximations, the following lower bound of λ_F is given, which is called Sidelnikov–Chabaud–Vaudenay’s bound.

1. When $n > m$ [Nyb91],

$$\lambda_F \geq 2^{-\frac{n}{2}};$$

2. When $n \leq m$ [CV94],

$$\lambda_F \geq 2^{-n} \cdot \sqrt{3 \times 2^n - 2 - 2 \frac{(2^n - 1)(2^{n-1} - 1)}{2^m - 1}}.$$

The lower bound of linearity λ_F is related to the notion of (almost) bent functions. The (almost) bent function is a Boolean function that can reach the lower bound of linearity λ_F under different cases. In essence, (almost) bent functions are a special type of Boolean function with optimal linearity.

Differential-linear cryptanalysis [LH94] is a technique that combines differential cryptanalysis and linear cryptanalysis to construct a distinguisher. The main idea is to exploit the differential characteristics and the linear approximations observed in short rounds of the cipher to build a longer-round distinguisher. This approach has been widely used [BDK03, Leu16, BLT20, BBC⁺22] since the seminal work of Langford and Hellman. However, on the concise theory of differential-linear cryptanalysis, there had been no progress for almost 20 years until 2014. Thanks to the link between multidimensional differential cryptanalysis and multidimensional linear approximations [BN13], Blondeau et al. [BLN17] exactly expressed the bias of a differential-linear approximation by

a closed expression for the first time. Based on the work of Blondeau et al., the concept of differential-linear connectivity table [BDKW19] was proposed to further develop the differential-linear cryptanalysis.

Until recent years, progress in practical applications of multi-party computation (MPC), fully homomorphic encryption (FHE), and zero-knowledge proofs (ZKP) motivate people to explore symmetric-key cryptographic algorithms over prime fields $\mathbb{F}_p$ for large p , such as MiMC [AGR⁺16], HadesMiMC [GLR⁺20], GMiMC [AGP⁺19], Poseidon [GKR⁺21] and other primitives [AAB⁺20, BBC⁺23], as well as over rings, such as Elisabeth [CHMS22], Rubato [HKL⁺22]. Moreover, this motivates us to reconsider traditional cryptanalysis techniques (such as differential cryptanalysis, linear cryptanalysis) for symmetric-key primitives working on prime fields $\mathbb{F}_p$ with large p , as well as on rings. Although differential cryptanalysis (through the theory of Markov ciphers) can be specified over an arbitrary group, linear cryptanalysis is based on a metric (the linear probability) that sticks to bit strings. It is highly required to generalize this notion when applying linear cryptanalysis to nonbinary block ciphers.

In [BSV07], Baignères et al. generalized the notion of linear distinguisher to arbitrary sets, and got sharp estimates of the data complexity and cumulate characteristics in linear hulls. In their work, they used group characters: for the function $F : \mathcal{G} \rightarrow \mathcal{H}$ between finite Abelian groups $\mathcal{G}$ and $\mathcal{H}$, the “linear approximation” of F corresponds to a pair of group characters (ψ_1, ψ_2) of $\mathcal{H}$ and $\mathcal{G}$. Namely, the correlation of the “linear approximation” (ψ_1, ψ_2) is equal to

$$C_{\psi_1, \psi_2}^F = \frac{1}{|\mathcal{G}|} \sum_{x \in \mathcal{G}} \psi_1(F(x)) \overline{\psi_2(x)},$$

where group characters ψ_1 and ψ_2 are group homomorphism $\psi_1 : \mathcal{H} \rightarrow \mathbb{C}^\times$ and a group homomorphism $\psi_2 : \mathcal{G} \rightarrow \mathbb{C}^\times$, respectively, and $\overline{\psi_1}$ denotes the complex conjugate of ψ_1 . For example, for $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, the linear approximation of F is

$$C_{u,v}^F = \frac{1}{2^n} \sum_{x \in \mathbb{F}_2^n} (-1)^{u \cdot F(x)} (-1)^{v \cdot x},$$

which is exactly the case in [Mat93]. For $F : \mathbb{Z}/N_1\mathbb{Z} \rightarrow \mathbb{Z}/N_2\mathbb{Z}$, the linear approximation of F is

$$C_{n_2, n_1}^F = \frac{1}{N_1} \sum_{x \in \mathbb{Z}/N_1\mathbb{Z}} e^{\frac{2n_2\pi i}{N_2} F(x)} e^{-\frac{2n_1\pi i}{N_1} x}.$$

For $F : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^m$, the linear approximation of F is

$$C_{u,v}^F = \frac{1}{p^n} \sum_{x \in \mathbb{F}_p^n} e^{\frac{2\pi i}{p} (F(x) \cdot u)} e^{-\frac{2\pi i}{p} (x \cdot v)},$$

where $x \cdot v$ is the inner product of $x, v \in \mathbb{F}_p^n$.

At CRYPTO 2021, Beyne [Bey21b] applied the linear approximation over finite Abelian groups to the cryptanalysis of FF3-1 [Dwo] and obtained a practical attack. Later at ASIACRYPT 2021 [Bey21a], by using the notion of correlation matrix of linear approximation, which is the same as the notion in [DGV94], Beyne showed the underlying link between the general linear approximation and many common cryptanalysis, such as invariant subspace attacks, nonlinear invariants and integral attacks. And for the symmetric-key cryptographic algorithms over prime fields, Xu et al. [XCWW23] applied it to the cryptanalysis of round-reduced HADESMiMC-128 [GLR⁺20]. In addition, following the notion of Baignères et al. [BSV07], Xu et al. [XCWW23] further proposed the differential-linear cryptanalysis over $\mathbb{F}_p^n$. Based on the general differential-linear cryptanalysis, they gave the first key recovery attack for round-reduced HADESMiMC-128 [GLR⁺20].

However, there remain some open problems that need to be addressed:

1. The lower bound of the linearity λ_F of the function $F : \mathcal{G} \rightarrow \mathcal{H}$ over an arbitrary finite Abelian group was first given by Pott in [Pot04]. Pott generalized the main results on nonlinear functions from the case of $\mathbb{F}_2^n$ to those on arbitrary Abelian groups by using the discrete Fourier transform instead of the Walsh–Hadamard transforms (the main tool to investigate Boolean functions on $\mathbb{F}_2^n$) and gave a lower bound of linearity λ_F . In fact, Pott only generalized the lower bound of the linearity λ_F , as established in [Nyb91] for $\mathbb{F}_2^n$ (one of the cases of Sidelnikov–Chabaud–Vaudenay’s bound), from $\mathbb{F}_2^n$ to finite Abelian groups, rather than achieving a complete generalization. Considering the new symmetric key cryptographic algorithms [AGR⁺16, AGP⁺19, GLR⁺20, GKR⁺21, GKRS22, GOPS22] over nonbinary fields or rings, it will greatly enhance the understanding of security for these ciphers if we could further completely generalize the Sidelnikov–Chabaud–Vaudenay’s bound [CV94] from $\mathbb{F}_2^n$ to finite Abelian groups.
2. Although the authors of [XCWW23] gave the expression of the correlation of the general differential-linear approximation, it relies on some assumptions which may not hold in practice. Therefore, for the differential-linear approximation over $\mathbb{F}_p^n$, the exact formula of the correlation needs to be derived under the independent assumption, as well as the differential-linear approximation over finite Abelian groups, i.e., the results in [BLN17] need to be generalized from $\mathbb{F}_2^n$ to $\mathbb{F}_p^n$ or finite Abelian groups.

Our Contribution. In this paper, we first derive the link between the differential cryptanalysis and linear approximations, which is some equivalence between differential probability and linear approximation. Owing to the link, we are able to further solve the two aforementioned problems.

As the first application, we give general results of linear approximations over finite Abelian groups, thereby generalizing the results in [Pot04] by Pott. Like [Nyb91, CV94], for a function $F : \mathcal{G} \rightarrow \mathcal{H}$ mapping from finite Abelian group $\mathcal{G}$ to $\mathcal{H}$, we use the “maximum value” of all non-trivial “linear approximation” λ_F ,

which is defined by

$$\lambda_F = \max_{\psi_1 \in \hat{\mathcal{H}}; \psi_2 \neq \hat{1}, \psi_1 \in \hat{\mathcal{G}}} \|C_{\psi_1, \psi_2}^F\|,$$

as the indicators to measure the resistance of a cipher to linear cryptanalysis. By the link between the differential cryptanalysis and linear approximation we establish, we are able to further deduce results of linear approximations over finite Abelian groups and completely generalize the Sidelnikov-Chabaud-Vaudenay's bound [CV94] from $\mathbb{F}_2^n$ to finite Abelian groups.

As the second application, we exactly derive the closed formula for the correlation of the differential-linear approximation over $\mathbb{F}_q^n$ [XCWW23] and finite Abelian groups, under the sole assumption that the two parts of the cipher are independent of each other.

Organization. The remainder of the paper is organized as follows: Section 2 includes the notations and preliminaries, respectively. Section 3 establishes new links between differential cryptanalysis and linear cryptanalysis over finite Abelian groups. Section 4 deduces a tighter lower bound of the linearity λ_F and completely generalizes the Sidelnikov-Chabaud-Vaudenay's bound [CV94] from $\mathbb{F}_2^n$ to finite Abelian groups. Section 5 gives the exact formula of differential-linear approximations over finite Abelian groups. The paper is concluded in Section 6.

2 Notations and Preliminaries

2.1 Notations

Let $\mathbb{F}_2 = \{0, 1\}$ be the binary field and $\mathbb{F}_2^n = \{0, 1\}^n$. Clearly, the set of all n -bit binary strings $\mathbb{F}_2^n$ with the exclusive-or operation $\oplus$ forms an Abelian group. For two vectors x and y in $\mathbb{F}_2^n$, the inner product of x and y is denoted by $x \cdot y$. Let $\mathbb{C}$ be the set of complex numbers. Then, the set of nonzero complex numbers $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ with the multiplication operation forms an Abelian group. For $z = a + bi \in \mathbb{C}$ with a and b being real numbers, its conjugate is denoted by $\bar{z} = a - bi$. The norm of z is defined as $\|z\| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$. For a finite set $\mathcal{G}$, the number of elements in $\mathcal{G}$ is denoted by $|\mathcal{G}|$. For integers a and b , a dividing b is denoted by $a \text{ divide } b$, a not dividing b is denoted by $a \nmid b$. For a real number x , $\lfloor x \rfloor$ is defined as the largest integer that does not exceed x .

2.2 Group Characters

Definition 1. Let $(\mathcal{G}_1, +)$ and $(\mathcal{G}_2, *)$ be two groups, if the function $f : \mathcal{G}_1 \rightarrow \mathcal{G}_2$ satisfies

$$f(g + h) = f(g) * f(h)$$

for all $g, h \in \mathcal{G}_1$, the function f is called a homomorphism. Moreover, if f is a bijection, we call it an isomorphism.

Definition 2 (Group Characters [Ter99]). Let $\mathcal{G}$ be a finite Abelian group. A group homomorphism $h : \mathcal{G} \rightarrow \mathbb{C}^*$ from $\mathcal{G}$ to $\mathbb{C}^*$ is called a group character of the finite Abelian group $\mathcal{G}$. We denote by $\hat{\mathcal{G}}$ the set of all characters of $\mathcal{G}$.

Example 1. The group homomorphism from $\mathbb{F}_2^n$ to $\mathbb{C}^*$ mapping $x \in \mathbb{F}_2^n$ to $(-1)^{a \cdot x}$ for some $a \in \mathbb{F}_2^n$ is a group character of $\mathbb{F}_2^n$. The group homomorphism from $\mathbb{Z}_n$ to $\mathbb{C}^*$ mapping $x \in \mathbb{Z}_n$ to $e^{-2\pi i a x/n}$ for some $a \in \mathbb{Z}_n$ is a group character of $\mathbb{Z}_n$.

It is easy to check that the unit circle $\{z \in \mathbb{C} : \|z\| = 1\} \subseteq \mathbb{C}$ is a group with the multiplication in $\mathbb{C}$. The following lemma tells us that the group characters of $\mathcal{G}$ are group homomorphisms from $\mathcal{G}$ to the unit circle in $\mathbb{C}$.

Lemma 1. Let $\mathcal{G}$ be a finite Abelian group. Then, for any $\chi \in \hat{\mathcal{G}}$ and $g \in \mathcal{G}$, $\|\chi(g)\| = 1$.

Proof. For $g \in \mathcal{G}$, there exists a positive integer N , such that $g^N = 1$. Then, for any $\chi \in \hat{\mathcal{G}}$, $\|\chi(g)\|^N = \|(\chi(g))^N\| = \|\chi(g^N)\| = 1$, and thus $\|\chi(g)\| = 1$.

Then, for $\psi \in \hat{\mathcal{G}}$ and $\varphi \in \hat{\mathcal{G}}$, the function mapping $x \in \mathcal{G}$ to $\psi(x)\varphi(x) \in \mathbb{C}^*$ is the group homomorphism from $\mathcal{G}$ to $\mathbb{C}^*$ due to Definition 2. We denote the operator $\star$ over $\hat{\mathcal{G}}$ such that for ψ and φ in $\hat{\mathcal{G}}$, $\psi \star \varphi \in \hat{\mathcal{G}}$ maps $x \in \mathcal{G}$ to $\psi(x)\varphi(x) \in \mathbb{C}^*$.

Definition 3. Let $\mathbf{1}_{\mathcal{G}} : \mathcal{G} \rightarrow \mathbb{C}^*$ be the homomorphism from the finite Abelian group $\mathcal{G}$ to $\mathbb{C}^*$ such that for all $g \in \mathcal{G}$, $\mathbf{1}_{\mathcal{G}}(g) = 1$. Then, $\mathbf{1}_{\mathcal{G}}$ is a trivial group character of $\mathcal{G}$.

It is easy to check that $\hat{\mathcal{G}}$ is also a finite Abelian group with the operation $\star$, where $\mathbf{1}_{\mathcal{G}} \in \hat{\mathcal{G}}$ is the identity element. The inverse element of ψ is denoted by ψ^{-1} , that is, $\psi \star \psi^{-1} = \psi^{-1} \star \psi = \mathbf{1}_{\mathcal{G}} \in \hat{\mathcal{G}}$. Note that the readers should not confuse ψ^{-1} with the inverse function of ψ , which may not exist at all. Let $\bar{\psi}$ be a function from $\mathcal{G}$ to $\mathbb{C}$ which maps $x \in \mathcal{G}$ to $\bar{\psi}(x)$, where $\bar{z}$ is conjugate of z for $z \in \mathbb{C}$. Then, it is easy to check that $\bar{\psi}$ is also a group character.

Lemma 2 ([Ter99]). The finite Abelian group $\hat{\mathcal{G}}$ is isomorphic to $\mathcal{G}$, which is denoted by $\hat{\mathcal{G}} \cong \mathcal{G}$. Therefore, $\hat{\hat{\mathcal{G}}} \cong \hat{\mathcal{G}} \cong \mathcal{G}$.

Lemma 3 ([Ter99]). Let $\mathcal{G}, \mathcal{H}$ be two finite Abelian groups. Then χ is a character of $(\mathcal{G}, \mathcal{H})$ if and only if $\chi = \chi_1 \times \chi_2$ for some $\chi_1 \in \hat{\mathcal{G}}$ and $\chi_2 \in \hat{\mathcal{H}}$, where $\times$ is the multiplication over $\mathbb{C}^*$, $(\mathcal{G}, \mathcal{H})$ is the direct product of $\mathcal{G}$ and $\mathcal{H}$.

Lemma 4 ([Ter99]). Different characters of $\mathcal{G}$ are “orthogonal”, namely, for $\chi, \psi \in \hat{\mathcal{G}}$,

$$\sum_{x \in \mathcal{G}} \overline{\chi(x)} \psi(x) = |\mathcal{G}| \cdot \Delta_{\chi, \psi}, \quad (1)$$

$$\text{where } \Delta_{\chi, \psi} = \begin{cases} 1, & \chi = \psi; \\ 0, & \chi \neq \psi. \end{cases}$$

Lemma 5. *Let χ be a group character for the finite Abelian group $\mathcal{G}$. Then, $\bar{\chi}(g) = \chi^{-1}(g)$ for all $g \in \mathcal{G}$.*

Proof. According to the definition of “ $\star$ ”, $\mathbf{1}_{\mathcal{G}}(g) = \chi \star (\chi^{-1})(g) = \chi(g)\chi^{-1}(g) = 1$. In addition, it follows from Lemma 1 that $\|\chi(g)\|^2 = \chi(g)\bar{\chi}(g) = 1$. Thus, we have $\chi(g)\bar{\chi}(g) = \chi(g)\chi^{-1}(g)$, which in turn implies that $\bar{\chi}(g) = \chi^{-1}(g)$.

2.3 Fourier Transformations over Finite Abelian Groups

Definition 4 (Fourier Transformation). *Let $f : \mathcal{G} \rightarrow \mathbb{C}$ be a function from a finite Abelian group $\mathcal{G}$ to $\mathbb{C}$. Then, the function $\hat{f} : \hat{\mathcal{G}} \rightarrow \mathbb{C}$ defined as $\hat{f} : \chi \mapsto \sum_{x \in \mathcal{G}} f(x)\chi(x)$ is called a Fourier transformation of f .*

Example 2. Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ be a Boolean function and $\chi \in \widehat{\mathbb{F}_2^n}$ be a group character such that $\chi(x) = (-1)^{a \cdot x}$ for some $a \in \mathbb{F}_2^n$. Then,

$$\hat{f}(\chi) = \sum_{x \in \mathbb{F}_2^n} f(x)\chi(x) = \sum_{x \in \mathbb{F}_2^n} f(x)(-1)^{a \cdot x},$$

which corresponds to the Walsh-Hadamard transformation.

Definition 5 (Inverse Fourier Transformation). *Let $h : \hat{\mathcal{G}} \rightarrow \mathbb{C}$ be a function, where $\mathcal{G}$ is a finite Abelian group. Then, the function $\tilde{h} : \mathcal{G} \rightarrow \mathbb{C}$ defined as $\tilde{h} : x \mapsto \frac{1}{|\mathcal{G}|} \sum_{\chi \in \hat{\mathcal{G}}} h(\chi)\bar{\chi}(x)$ is called the inverse Fourier transformation of h .*

Lemma 6. *Let $f : \mathcal{G} \rightarrow \mathbb{C}$ be a function from a finite Abelian group $\mathcal{G}$ to $\mathbb{C}$. Then, $\tilde{\hat{f}} = f$.*

Next, we define the convolution operator “ $\ast$ ”, and we remark that “ $\star$ ” and “ $\ast$ ” are different operators in this paper.

Definition 6 (Convolution). *Let f and g be functions from $(\mathcal{G}, +)$ to $\mathbb{C}$. Then, the convolution $f \ast g$ of f and g is defined as*

$$(f \ast g)(x) = \sum_{y \in \mathcal{G}} f(x - y)g(y).$$

Lemma 7. *Let f and g be both functions from $(\mathcal{G}, +)$ to $\mathbb{C}$. Then, $\widehat{f \ast g}(\chi) = \hat{f}(\chi)\hat{g}(\chi)$.*

2.4 Differential Probability over Abelian Groups

For convenience, for the group operation of finite Abelian groups $\mathcal{G}$, we denote it by $+$ and omit writing $+\mathcal{G}$.

Definition 7. Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. The probability of the differential of F with input difference $\alpha \in \mathcal{G}$ and output difference $\beta \in \mathcal{H}$ is defined as

$$\Pr[\alpha \xrightarrow{F} \beta] = \frac{\delta_F(\alpha, \beta)}{|\mathcal{G}|},$$

where $\delta_F(\alpha, \beta) = |\{x \in \mathcal{G} | F(x) - F(x - \alpha) = \beta\}|$.

Example 3. For any function $F : \mathcal{G} \rightarrow \mathcal{H}$, $\delta_F(0, 0) = |\mathcal{G}|$.

Example 4. Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be a function. For $(\alpha, \beta) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$, $\Pr[\alpha \xrightarrow{F} \beta] = 2^{-n} \delta_F(\alpha, \beta)$, where $\delta_F(\alpha, \beta) = |\{x \in \mathbb{F}_2^n | F(x) \oplus F(x \oplus \alpha) = \beta\}|$.

Example 5. Let $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function with $\mathcal{G} = \mathcal{H} = (\mathbb{Z}_{2^n}, +)$. For $\alpha, \beta \in \mathbb{Z}_{2^n}$, we have

$$\Pr[\alpha \xrightarrow{F} \beta] = \frac{\delta_F(\alpha, \beta)}{2^n},$$

where $\delta_F(\alpha, \beta) = |\{x \in \mathcal{G} | F(x) - F(x - \alpha) = \beta\}|$.

Let $\mathcal{S} \subseteq \mathbb{Z}$ be a set of integers. We use $\min^+(\mathcal{S})$ to denote the minimum positive number in $\mathcal{S}$. For example, $\min^+(\mathcal{S}) = 2$ for $\mathcal{S} = \{0, 3, 5, 5, 3, 0, 2\}$.

Definition 8. For a set $\mathcal{F}$ of functions from a finite Abelian group $(\mathcal{G}, +)$ to another finite Abelian group $(\mathcal{H}, +)$, we denote

$$\rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}} = \min^+\{\delta_F(\alpha, \beta) : F \in \mathcal{F}, \alpha \in \mathcal{G}, \beta \in \mathcal{H}\}.$$

When $\mathcal{F}$ is the set that consists of all the functions from a finite Abelian group $(\mathcal{G}, +)$ to another finite Abelian group $(\mathcal{H}, +)$, we may omit the superscript for simplicity. Or when $\mathcal{G}$, $\mathcal{H}$ and $\mathcal{F}$ are clear from the context, we may omit the superscript and subscript for simplicity.

Example 6. Let $\mathcal{F}$ be the set of all vectorial Boolean functions from $\mathbb{F}_2^n$ to $\mathbb{F}_2^n$. Then, due to the self-invertibility of $\oplus$, we have $\rho_{\mathbb{F}_2^n, \mathbb{F}_2^n}^{\mathcal{F}} \geq 2$. For instance, for $(x_1, x_2, \dots, x_n) \in \mathbb{F}_2^n$, the vectorial Boolean functions $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is defined as

$$(x_1, x_2, \dots, x_n) \mapsto (x_1 \wedge x_n, x_2 \wedge x_1, \dots, x_n \wedge x_{n-1}).$$

If we let $\alpha = (1, 1, 1, \dots, 1)$ and $\beta = (1, 1, 0, \dots, 0)$, we have $\delta_f(\alpha, \beta) = 2$ (see [KLT15] Theorem 2). Thus, $\rho_{\mathbb{F}_2^n, \mathbb{F}_2^n}^{\mathcal{F}} = 2$.

Example 7. Let $\mathcal{F}$ be the set of all functions from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^n}$. Firstly, due to the self-invertibility of $+$ on the $\mathbb{F}_{2^n}$, we have $\rho_{\mathbb{F}_{2^n}, \mathbb{F}_{2^n}}^{\mathcal{F}} \geq 2$. Because $(\mathbb{F}_{2^n}, +)$ is isomorphic to $(\mathbb{F}_2^n, +)$, we have $\rho_{\mathbb{F}_{2^n}, \mathbb{F}_{2^n}}^{\mathcal{F}} = 2$ according to the Example 6.

2.5 Correlation of Linear Approximations over Abelian Groups

Definition 9 ([Bey21b, BSV07]). Let $\mathcal{G}$ and $\mathcal{H}$ be two finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. The correlation of the linear approximation of F with the input character $\psi \in \hat{\mathcal{G}}$ and the output character $\chi \in \hat{\mathcal{H}}$ is defined as

$$\mathcal{C}_{\psi, \chi}^F = \frac{1}{|\mathcal{G}|} \sum_{x \in \mathcal{G}} \overline{\psi}(x) \chi(F(x)).$$

Example 8. The correlation $\mathcal{C}_{u, v}^F$ of the linear approximation of a vectorial Boolean function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ with input mask $u \in \mathbb{F}_2^n$ (corresponding to the character $x \mapsto (-1)^{u \cdot x}$) and output mask $v \in \mathbb{F}_2^n$ (corresponding to the character $x \mapsto (-1)^{v \cdot x}$) is defined as

$$\mathcal{C}_{u, v}^F = \frac{1}{2^n} \sum_{x \in \mathbb{F}_2^n} (-1)^{u \cdot x} (-1)^{v \cdot F(x)} = \frac{1}{2^n} \sum_{x \in \mathbb{F}_2^n} (-1)^{u \cdot x \oplus v \cdot F(x)}.$$

Example 9 ([BSV07, Bey21b]). The correlation $\mathcal{C}_{a, b}^F$ of the linear approximation of a function $F : \mathbb{Z}_n \rightarrow \mathbb{Z}_m$ with the input mask $a \in \mathbb{Z}_n$ (corresponding to the character $x \mapsto e^{\frac{2\pi i a x}{n}}$) and the output mask $b \in \mathbb{Z}_m$ (corresponding to the character $x \mapsto e^{\frac{2\pi i b x}{m}}$) is defined as

$$\mathcal{C}_{a, b}^F = \frac{1}{n} \sum_{x \in \mathbb{Z}_n} e^{\frac{-2\pi i a x}{n}} e^{\frac{2\pi i b F(x)}{m}}.$$

Example 10 ([Bey21b]). The correlation $\mathcal{C}_{a, b}^\boxplus$ of the modulo addition $\boxplus$ with the input mask $(a, b) \in (\mathbb{Z}_{2^n}, \mathbb{Z}_{2^n})$ (corresponding to the character $(x, y) \mapsto e^{\frac{2\pi i a x}{2^n}} e^{\frac{2\pi i b y}{2^n}}$) and the output mask $c \in \mathbb{Z}_{2^n}$ (corresponding to the character $x \mapsto e^{\frac{2\pi i c x}{2^n}}$) can be calculated as

$$\begin{aligned} \mathcal{C}_{(a, b), c}^F &= 2^{-2n} \sum_{x, y \in \mathbb{Z}_{2^n}} e^{\frac{2\pi i a x}{2^n}} e^{\frac{2\pi i b y}{2^n}} e^{\frac{-2\pi i c (x \boxplus y)}{2^n}} \\ &= 2^{-2n} \sum_{x, y \in \mathbb{Z}_{2^n}} e^{\frac{2\pi i a x}{2^n}} e^{\frac{2\pi i b y}{2^n}} e^{\frac{-2\pi i c x}{2^n}} e^{\frac{-2\pi i c y}{2^n}} \\ &= 2^{-n} \sum_{x \in \mathbb{Z}_{2^n}} e^{\frac{2\pi i (a-c)x}{2^n}} 2^{-n} \sum_{y \in \mathbb{Z}_{2^n}} e^{\frac{2\pi i (b-c)y}{2^n}} \\ &= \Delta_{a, c} \Delta_{b, c} = \begin{cases} 1, & a = b = c; \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

For the correlation of linear approximations over a finite Abelian group, we have the following property.

Lemma 8. Let ψ and χ be characters of the finite Abelian groups $\mathcal{G}$ and $\mathcal{H}$, respectively, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. Then, $\overline{\mathcal{C}_{\psi, \chi}^F} = \mathcal{C}_{\psi^{-1}, \chi^{-1}}^F$.

Proof. According to Definition 9 and Lemma 5, we have

$$\begin{aligned}\overline{\mathcal{C}_{\psi,\chi}^F} &= \frac{1}{|\mathcal{G}|} \sum_{x \in \mathcal{G}} \overline{\psi(x) \chi(F(x))} = \frac{1}{|\mathcal{G}|} \sum_{x \in \mathcal{G}} \overline{\psi(x)} \overline{\chi(F(x))} \\ &= \frac{1}{|\mathcal{G}|} \sum_{x \in \mathcal{G}} \overline{\psi^{-1}(x)} \chi^{-1}(F(x)) = \mathcal{C}_{\psi^{-1}, \chi^{-1}}^F.\end{aligned}$$

3 The Link between the Differential Cryptanalysis and Linear Cryptanalysis and Linearity over Finite Abelian Groups

In this section, we establish new links between differential cryptanalysis and linear cryptanalysis over finite Abelian groups, which will be used in Section 4 to deduce a tighter lower bound of λ_F . We first define two special index functions θ_F and θ_F^* , which will be used to prove Lemma 11.

Definition 10. Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. The functions $\theta_F : (\mathcal{G}, \mathcal{H}) \rightarrow \mathbb{R}$ and $\theta_F^* : (\mathcal{G}, \mathcal{H}) \rightarrow \mathbb{R}$ are defined as

$$\theta_F(a, b) = \begin{cases} 1, & b = F(a) \\ 0, & b \neq F(a) \end{cases}, \quad \theta_F^*(a, b) = \begin{cases} 1, & b = -F(-a) \\ 0, & b \neq -F(-a) \end{cases},$$

respectively.

For θ_F and θ_F^* , we have the following property.

Lemma 9. $\hat{\theta}_F(\psi, \chi) = |\mathcal{G}| \mathcal{C}_{\psi^{-1}, \chi}$, and $\hat{\theta}_F^*(\psi, \chi) = \hat{\theta}_F(\psi^{-1}, \chi^{-1})$.

Proof. For $\hat{\theta}_F(\psi, \chi)$, we have

$$\begin{aligned}\hat{\theta}_F(\psi, \chi) &= \sum_{(a,b) \in (\mathcal{G}, \mathcal{H})} \chi(b) \psi(a) \theta_F(a, b) = \sum_{a \in \mathcal{G}} \chi(F(a)) \psi(a) \\ &= \sum_{a \in \mathcal{G}} \chi(F(a)) \overline{\overline{\psi(a)}} = \sum_{a \in \mathcal{G}} \chi(F(a)) \overline{\psi^{-1}(a)} \\ &= |\mathcal{G}| \mathcal{C}_{\psi^{-1}, \chi}.\end{aligned}$$

For $\hat{\theta}_F^*(\psi, \chi)$, we have

$$\begin{aligned}\hat{\theta}_F^*(\psi, \chi) &= \sum_{(a,b) \in (\mathcal{G}, \mathcal{H})} \chi(b) \psi(a) \theta_F^*(a, b) \\ &= \sum_{a \in \mathcal{G}} \chi(-F(-a)) \psi(a).\end{aligned}$$

Since for any $g \in \mathcal{G}$ and $h \in \mathcal{H}$, $\chi(g)\chi(-g) = 1$ and $\psi(h)\psi(-h) = 1$, $\chi(-g) = \chi^{-1}(g)$, and $\psi^{-1}(h) = \psi(-h)$. Consequently, we have

$$\begin{aligned}\hat{\theta}_F^*(\psi, \chi) &= \sum_{a \in \mathcal{G}} \chi(-F(-a))\psi(-(-a)) \\ &= \sum_{-a \in \mathcal{G}} \chi^{-1}(F(-a))\psi^{-1}(-a) = \hat{\theta}_F(\psi^{-1}, \chi^{-1}).\end{aligned}$$

Then, we have the following relationship between $(\theta_F^* * \theta_F)(\alpha, \beta)$ and $\delta_F(\alpha, \beta)$.

Lemma 10. $(\theta_F^* * \theta_F)(\alpha, \beta) = \delta_F(\alpha, \beta)$.

Proof. According to Definition 6, we have

$$\begin{aligned}(\theta_F^* * \theta_F)(\alpha, \beta) &= \sum_{(x, y) \in (\mathcal{G}, \mathcal{H})} \theta_F^*(\alpha - x, \beta - y)\theta_F(x, y) \\ &= \sum_{x \in \mathcal{G}} \theta_F^*(\alpha - x, \beta - F(x)) \\ &= |\{x \in \mathcal{G} : \beta - F(x) = -F(-(\alpha - x))\}| \\ &= |\{x \in \mathcal{G} : F(x) - F(x - \alpha) = \beta\}| \\ &= \delta_F(\alpha, \beta).\end{aligned}$$

Then, we give the first relationship between the differential probability and the correlation of linear approximations.

Lemma 11. *Let $\mathcal{G}$ and $\mathcal{H}$ be both finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. Then, we have*

$$\delta_F(\alpha, \beta) = \frac{|\mathcal{G}|}{|\mathcal{H}|} \sum_{\chi \in \hat{\mathcal{H}}, \psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha)\chi(\beta)\mathcal{C}_{\psi, \chi}^F \overline{\mathcal{C}_{\psi, \chi}^F}, \quad (2)$$

or equivalently,

$$\mathcal{C}_{\psi, \chi}^F \overline{\mathcal{C}_{\psi, \chi}^F} = \frac{1}{|\mathcal{G}|^2} \sum_{\alpha \in \mathcal{G}, \beta \in \mathcal{H}} \psi(\alpha)\bar{\chi}(\beta)\delta_F(\alpha, \beta).$$

Proof. According to Lemma 10, we have

$$\begin{aligned}
\delta_F(\alpha, \beta) &= (\theta_F^* * \theta_F)(\alpha, \beta) \\
&= \widetilde{(\theta_F^* * \theta_F)}(\alpha, \beta) = \widetilde{(\theta_F^* \times \hat{\theta}_F)}(\alpha, \beta) \quad (\text{Lemma 7}) \\
&= \frac{1}{|\mathcal{G}|} \frac{1}{|\mathcal{H}|} \sum_{\psi \in \hat{\mathcal{G}}, \lambda \in \hat{\mathcal{H}}} \overline{\psi(\alpha)} \overline{\lambda(\beta)} \hat{\theta}_F^*(\psi, \lambda) \hat{\theta}_F(\psi, \lambda) \quad (\text{Definition 5}) \\
&= \frac{|\mathcal{G}|}{|\mathcal{H}|} \sum_{\psi \in \hat{\mathcal{G}}, \lambda \in \hat{\mathcal{H}}} \overline{\psi(\alpha)} \overline{\lambda(\beta)} \mathcal{C}_{\psi, \lambda^{-1}} \mathcal{C}_{\psi^{-1}, \lambda} \quad (\text{Lemma 10}) \\
&= \frac{|\mathcal{G}|}{|\mathcal{H}|} \sum_{\psi \in \hat{\mathcal{G}}, \chi \in \hat{\mathcal{H}}} \overline{\psi(\alpha)} \chi(\beta) \mathcal{C}_{\psi, \chi} \mathcal{C}_{\psi^{-1}, \chi^{-1}} \\
&= \frac{|\mathcal{G}|}{|\mathcal{H}|} \sum_{\psi \in \hat{\mathcal{G}}, \chi \in \hat{\mathcal{H}}} \overline{\psi(\alpha)} \chi(\beta) \mathcal{C}_{\psi, \chi} \overline{\mathcal{C}_{\psi, \chi}}. \quad (\text{Lemma 8})
\end{aligned}$$

Equivalently, we can see that

$$\begin{aligned}
&\sum_{\alpha \in \mathcal{G}, \beta \in \mathcal{H}} \psi(\alpha) \overline{\chi(\beta)} \delta_F(\alpha, \beta) \\
&= \frac{|\mathcal{G}|^2}{|\mathcal{G}| |\mathcal{H}|} \sum_{\alpha \in \mathcal{G}, \beta \in \mathcal{H}} \psi(\alpha) \overline{\chi(\beta)} \sum_{\sigma \in \hat{\mathcal{G}}, \tau \in \hat{\mathcal{H}}} \overline{\sigma(\alpha)} \tau(\beta) \mathcal{C}_{\sigma, \tau} \overline{\mathcal{C}_{\sigma, \tau}} \quad (\text{Equation (2)}) \\
&= \frac{|\mathcal{G}|^2}{|\mathcal{G}| |\mathcal{H}|} \sum_{\sigma \in \hat{\mathcal{G}}, \tau \in \hat{\mathcal{H}}} \mathcal{C}_{\sigma, \tau} \overline{\mathcal{C}_{\sigma, \tau}} \sum_{\alpha \in \mathcal{G}} \psi(\alpha) \overline{\sigma(\alpha)} \sum_{\beta \in \mathcal{H}} \tau(\beta) \overline{\chi(\beta)} = |\mathcal{G}|^2 \mathcal{C}_{\psi, \chi}^F \overline{\mathcal{C}_{\psi, \chi}^F}.
\end{aligned}$$

Gathering Lemma 9, Lemma 10 and Lemma 11, we finally draw the first main result in Theorem 1, which gives the second relationship between the differential probability and the correlation of linear approximations.

Theorem 1. *Let $\mathcal{G}$ and $\mathcal{H}$ be both finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. Then, we have*

$$\sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}, \chi \neq 1} \|\mathcal{C}_{\psi, \chi}^F\|^4 = \frac{|\mathcal{H}|}{|\mathcal{G}|^3} \sum_{\alpha \in \mathcal{G}, \alpha \neq 0} \sum_{\beta \in \mathcal{H}} \delta_F(\alpha, \beta)^2 + \frac{|\mathcal{H}|}{|\mathcal{G}|} - 1.$$

In addition, we call $\sum_{\alpha \in \mathcal{G}, \alpha \neq 0} \sum_{\beta \in \mathcal{H}} \delta_F(\alpha, \beta)^2$ as the square sum of differential probability.

Proof. Direct application of Definition 4 and Lemma 11 would give

$$\frac{1}{|\mathcal{G}|^2} \hat{\delta}_F(\psi, \chi) = \|\mathcal{C}_{\psi, \chi}^F\|^2 = \mathcal{C}_{\psi, \chi}^F \overline{\mathcal{C}_{\psi, \chi}^F}.$$

Therefore, we have

$$\begin{aligned}
& \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}, \chi \neq \mathbf{1}_{\mathcal{H}}} \|\mathcal{C}_{\psi, \chi}^F\|^4 \\
&= \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}} \|\mathcal{C}_{\psi, \chi}^F\|^4 - \sum_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \mathbf{1}}^F\|^4 \\
&= \frac{1}{|\mathcal{G}|^4} \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}} (\widehat{\delta_F}(\psi, \bar{\chi}))^2 - \sum_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \mathbf{1}}^F\|^4 \quad (\text{Lemma 11}) \\
&= \frac{1}{|\mathcal{G}|^4} \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}} \widehat{\delta_F * \delta_F}(\psi, \bar{\chi}) - \sum_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \mathbf{1}}^F\|^4 \quad (\text{Lemma 7}) \\
&= \frac{1}{|\mathcal{G}|^4} \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}} \widehat{\delta_F * \delta_F}(\psi, \chi) - \sum_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \mathbf{1}}^F\|^4 \\
&= \frac{|\mathcal{H}|}{|\mathcal{G}|^3} \frac{1}{|\mathcal{G}||\mathcal{H}|} \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}} \bar{\psi}(0) \bar{\chi}(0) \widehat{\delta_F * \delta_F}(\psi, \chi) - \sum_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \mathbf{1}}^F\|^4 \\
&= \frac{|\mathcal{H}|}{|\mathcal{G}|^3} (\delta_F * \delta_F)(0, 0) - \sum_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \mathbf{1}}^F\|^4. \quad (\text{Lemma 6})
\end{aligned}$$

Due to the fact that $\mathcal{C}_{\psi, \mathbf{1}_{\mathcal{H}}}^F = \begin{cases} 1, & \psi = \mathbf{1}_{\mathcal{G}} \\ 0, & \text{otherwise} \end{cases}$, it follows that

$$\sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}, \chi \neq \mathbf{1}_{\mathcal{H}}} \|\mathcal{C}_{\psi, \chi}^F\|^4 = \frac{|\mathcal{H}|}{|\mathcal{G}|^3} \sum_{\alpha \in \mathcal{G}, \alpha \neq 0} \sum_{\beta \in \mathcal{H}} \delta_F(\alpha, \beta)^2 + \frac{|\mathcal{H}|}{|\mathcal{G}|} - 1.$$

Example 11. Applying Theorem 1 to a vectorial Boolean function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, we have

$$\sum_{u \in \mathbb{F}_2^n} \sum_{v \in \mathbb{F}_2^m, v \neq 0} \|\mathcal{C}_{u, v}^F\|^4 = 2^{m-3n} \sum_{\substack{\alpha \in \mathbb{F}_2^n \\ \alpha \neq 0}} \sum_{\beta \in \mathbb{F}_2^m} \delta_F(\alpha, \beta)^2 + 2^{m-n} - 1.$$

It is worth noting that the result in this example is exactly the same as the one obtained in [CV94].

In the following, we introduce the definition of the linearity of functions to any finite Abelian groups and give its lower bound based on the Parseval equation, which is the corollary of Lemma 11. The main reason why we use Lemma 11 instead of other methods to conclude the Parseval equation is to show the inner link between the differential probability with zero input difference and the Parseval equation.

Definition 11 (Linearity over Abelian groups). Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. The nonlinearity λ_F of function F over finite Abelian groups is defined as

$$\lambda_F = \max_{(\psi, \chi) \in (\hat{\mathcal{G}}, \hat{\mathcal{H}}), \chi \neq \mathbf{1}_{\mathcal{H}}} \|\mathcal{C}_{\psi, \chi}\|. \quad (3)$$

Theorem 2 (Parseval equation over Abelian groups).

$$\sum_{\psi \in \hat{\mathcal{G}}} \mathcal{C}_{\psi, \chi} \overline{\mathcal{C}_{\psi, \chi}} = \sum_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \chi}\|^2 = 1. \quad (4)$$

Proof. For $\delta_F(\alpha, \beta)$, when $\beta \neq 0_{\mathcal{H}}$, $\delta_F(0_{\mathcal{G}}, \beta) = 0$ and $\delta_F(0_{\mathcal{G}}, 0_{\mathcal{H}}) = |\mathcal{G}|$. For any $\psi \in \hat{\mathcal{G}}$ and $\chi \in \hat{\mathcal{H}}$, we have $\psi(0_{\mathcal{G}}) = 1$, $\chi(0_{\mathcal{H}}) = 1$.

Let $\delta_F(0_{\mathcal{G}}, \beta) = f(\beta)$, for any $\chi \in \hat{\mathcal{H}}$, we then have

$$\hat{f}(\chi) = \sum_{g \in \mathcal{H}} \chi(g) \delta_F(0_{\mathcal{G}}, g) = \chi(0) \delta_F(0_{\mathcal{G}}, 0_{\mathcal{H}}) = |\mathcal{G}|.$$

From Lemma 11, it follows that

$$f(\beta) = \frac{|\mathcal{G}|}{|\mathcal{H}|} \sum_{\psi \in \hat{\mathcal{G}}, \chi \in \hat{\mathcal{H}}} \chi(\beta) \mathcal{C}_{\psi, \chi} \overline{\mathcal{C}_{\psi, \chi}}.$$

Then,

$$\begin{aligned} \hat{f}(\chi) &= \frac{|\mathcal{G}|}{|\mathcal{H}|} \sum_{g \in \mathcal{H}} \chi(g) \sum_{\chi_1 \in \hat{\mathcal{H}}} \sum_{\psi \in \hat{\mathcal{G}}} \chi_1(g) \mathcal{C}_{\psi, \chi_1} \overline{\mathcal{C}_{\psi, \chi_1}} \\ &= \frac{|\mathcal{G}|}{|\mathcal{H}|} \sum_{\chi_1 \in \hat{\mathcal{H}}} \sum_{\psi \in \hat{\mathcal{G}}} \mathcal{C}_{\psi, \chi_1} \overline{\mathcal{C}_{\psi, \chi_1}} \sum_{g \in \mathcal{H}} \chi(g) \chi_1(g) \\ &= |\mathcal{G}| \sum_{\chi_1 \in \hat{\mathcal{H}}} \sum_{\psi \in \hat{\mathcal{G}}} \mathcal{C}_{\psi, \chi_1} \overline{\mathcal{C}_{\psi, \chi_1}} \Delta_{\bar{\chi}, \chi_1} \quad (\text{Lemma 4}) \\ &= |\mathcal{G}| \sum_{\psi \in \hat{\mathcal{G}}} \mathcal{C}_{\psi, \bar{\chi}} \overline{\mathcal{C}_{\psi, \bar{\chi}}}. \end{aligned}$$

Thus,

$$\sum_{\psi \in \hat{\mathcal{G}}} \mathcal{C}_{\psi, \chi} \overline{\mathcal{C}_{\psi, \chi}} = \sum_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \chi}\|^2 = 1.$$

According to the Parseval equation, we have $\lambda_F \geq |\mathcal{G}|^{-\frac{1}{2}}$:

Corollary 1. $\lambda_F \geq |\mathcal{G}|^{-\frac{1}{2}}$.

Proof. Note that

$$\lambda_F = \max_{\chi \in \hat{\mathcal{H}}, \chi \neq \mathbf{1}} \max_{\psi \in \hat{\mathcal{G}}} (\|\mathcal{C}_{\psi, \chi}\|^2)^{\frac{1}{2}} = \max_{\chi \in \hat{\mathcal{H}}, \chi \neq \mathbf{1}} (\max_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \chi}\|^2)^{\frac{1}{2}}.$$

For $\max_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \chi}\|^2$, we then have

$$\max_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \chi}\|^2 \geq \frac{\sum_{\psi \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi, \chi}\|^2}{|\mathcal{G}|} = \frac{1}{|\mathcal{G}|}.$$

Thus, $\lambda_F \geq |\mathcal{G}|^{-\frac{1}{2}}$.

Remark 1. In Appendix A, we show that only when $\rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}} \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$, $|\mathcal{G}| \mid |\mathcal{H}|$, λ_F can reach $|\mathcal{G}|^{-\frac{1}{2}}$. For other cases, $\lambda_F > |\mathcal{G}|^{-\frac{1}{2}}$. Thus, the lower bound of the linearity λ_F from Corollary 1 is not tight.

4 A Tighter bound of λ_F

As the first application of the link between differential cryptanalysis and linear cryptanalysis, in this section we give a tighter lower bound of the linearity λ_F by gathering the results in Section 3 and complete the generalization of Sidelnikov-Chabaud-Vaudenay's bound from $\mathbb{F}_2^n$ to finite Abelian groups.

4.1 On The Lower Bound of The Square Sum of Differential Probability

Before we discuss the linearity, we consider $\sum_{\alpha \in \mathcal{G}, \alpha \neq 0} \sum_{\beta \in \mathcal{H}} \delta_F(\alpha, \beta)^2$ (the right part) in the equation in Theorem 1. We calculate its lower bound so that we can get the lower bound of the linear approximations (the left part), which will be directly used to deduce a tighter lower bound of λ_F .

Let $G = \{\alpha_1, \dots, \alpha_{|\mathcal{G}|-1}, \alpha_{|\mathcal{G}|}\}$ and $H = \{\beta_1, \dots, \beta_{|\mathcal{H}|-1}, \beta_{|\mathcal{H}|}\}$ with $a_{|\mathcal{G}|} = 0$. Then we have

$$\sum_{\alpha \in \mathcal{G}, \alpha \neq 0} \sum_{\beta \in \mathcal{H}} \delta_F(\alpha, \beta)^2 = \sum_{i=1}^{|\mathcal{G}|-1} \sum_{j=1}^{|\mathcal{H}|} \delta_F(\alpha_i, \beta_j)^2. \quad (5)$$

Let $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|})$, $\mathbb{D}_{\mathcal{G}, \mathcal{H}} = \{\mathbf{d} \in \mathbb{Z}^{|\mathcal{H}|} : d_i = 0 \text{ or } d_i \geq \rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}}, \text{ for } 0 \leq i \leq |\mathcal{H}|, \text{ and } \sum_{i=1}^{|\mathcal{H}|} d_i = |\mathcal{G}|\}$, and $\xi_{\mathcal{G}, \mathcal{H}} = \min_{\mathbf{d} \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}} \sum_{j=1}^{|\mathcal{H}|} (d_j)^2$. It is easy to see that for any $i \in \{1, 2, \dots, |\mathcal{G}|-1\}$, $\sum_{j=1}^{|\mathcal{H}|} \delta_F(\alpha_i, \beta_j) = |\mathcal{G}|$, and for $1 \leq j \leq |\mathcal{H}|$, $\delta_F(\alpha_i, \beta_j) = 0$ or $\rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}} \leq \delta_F(\alpha_i, \beta_j) \leq |\mathcal{G}|$. Therefore, for any $i \in \{1, \dots, |\mathcal{G}|-1\}$,

$$\sum_{j=1}^{|\mathcal{H}|} \delta_F(\alpha_i, \beta_j)^2 \geq \xi_{\mathcal{G}, \mathcal{H}}. \quad (6)$$

Consequently, $\sum_{i=1}^{|\mathcal{G}|-1} \sum_{j=1}^{|\mathcal{H}|} \delta_F(\alpha_i, \beta_j)^2$ is lower bounded by $(|\mathcal{G}|-1)\xi_{\mathcal{G}, \mathcal{H}}$. In the following, we will give the $\xi_{\mathcal{G}, \mathcal{H}}$ under different cases.

Definition 12. Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups, and

$$\zeta_{\mathcal{G}, \mathcal{H}} = \min_{\mathbf{d} \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}} \sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2, \quad (7)$$

where $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|})$ and

$$\mathbb{D}_{\mathcal{G}, \mathcal{H}} = \{\mathbf{d} \in \mathbb{Z}^{|\mathcal{H}|} : d_i = 0 \text{ or } d_i \geq \rho, \text{ for } 0 \leq i \leq |\mathcal{H}|, \text{ and } \sum_{i=1}^{|\mathcal{H}|} d_i = |\mathcal{G}|\}.$$

We can see the following relationship between $\xi_{\mathcal{G}, \mathcal{H}}$ and $\zeta_{\mathcal{G}, \mathcal{H}}$.

Lemma 12.

$$\zeta_{\mathcal{G}, \mathcal{H}} + \frac{|\mathcal{G}|^2}{|\mathcal{H}|} = \xi_{\mathcal{G}, \mathcal{H}}.$$

Proof. For $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|}) \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}$, we can see that

$$\begin{aligned} & \sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + \frac{|\mathcal{G}|^2}{|\mathcal{H}|} \\ &= \left(\sum_{i=1}^{|\mathcal{H}|} d_i^2 - 2 \frac{|\mathcal{G}|}{|\mathcal{H}|} \sum_{i=1}^{|\mathcal{H}|} d_i + \frac{|\mathcal{G}|^2}{|\mathcal{H}|^2} \sum_{i=1}^{|\mathcal{H}|} 1 \right) + \frac{|\mathcal{G}|^2}{|\mathcal{H}|} \\ &= \left(\sum_{i=1}^{|\mathcal{H}|} d_i^2 - 2 \frac{|\mathcal{G}|^2}{|\mathcal{H}|} + \frac{|\mathcal{G}|^2}{|\mathcal{H}|} \right) + \frac{|\mathcal{G}|^2}{|\mathcal{H}|} \\ &= \sum_{i=1}^{|\mathcal{H}|} d_i^2. \end{aligned}$$

Thus, we have

$$\zeta_{\mathcal{G}, \mathcal{H}} + \frac{|\mathcal{G}|^2}{|\mathcal{H}|} = \xi_{\mathcal{G}, \mathcal{H}}.$$

Obviously, according to Lemma 12, if we know the value of $\zeta_{\mathcal{G}, \mathcal{H}}$, then we can get $\xi_{\mathcal{G}, \mathcal{H}}$ immediately. Next, we will give the value of $\zeta_{\mathcal{G}, \mathcal{H}}$ under different cases, the divisibility of the size of group $\mathcal{H}$ by the size of $\mathcal{G}$, the fulfillment of the condition that ρ lies either below or above the floor of the ratio of the size of $\mathcal{G}$ and the size of $\mathcal{H}$.

Cases: $|\mathcal{H}|$ divide $|\mathcal{G}|$ and $\rho \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$

Lemma 13. *Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups with $|\mathcal{H}|$ dividing $|\mathcal{G}|$ and $\rho \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$. Then*

$$\zeta_{\mathcal{G}, \mathcal{H}} = 0.$$

Accordingly,

$$\xi_{\mathcal{G}, \mathcal{H}} = \frac{|\mathcal{G}|^2}{|\mathcal{H}|}.$$

Proof. For $d \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}$, we can see that $\sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \geq 0$. By setting $d' = \left(\frac{|\mathcal{G}|}{|\mathcal{H}|}, \dots, \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)$, we have $d' \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}$ and get $\zeta_{\mathcal{G}, \mathcal{H}} = 0$. According to Lemma 12, $\xi_{\mathcal{G}, \mathcal{H}} = \frac{|\mathcal{G}|^2}{|\mathcal{H}|}$.

Cases: $|\mathcal{H}| \nmid |\mathcal{G}|$ and $\rho \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$

Lemma 14. Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups with $|\mathcal{H}| \nmid |\mathcal{G}|$ and $\rho \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$. Then

$$\zeta_{\mathcal{G}, \mathcal{H}} = \min_{\mathbf{d} \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}} \sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 = \min_{\mathbf{d} \in \mathbb{B}_{\mathcal{G}, \mathcal{H}}} \sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2,$$

where $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|})$,

$$\mathbb{D}_{\mathcal{G}, \mathcal{H}} = \{ \mathbf{d} \in \mathbb{Z}^{|\mathcal{H}|} : d_i = 0 \text{ or } d_i \geq \rho, \text{ for } 0 \leq i \leq |\mathcal{H}|, \text{ and } \sum_{i=1}^{|\mathcal{H}|} d_i = |\mathcal{G}| \},$$

$$\mathbb{B}_{\mathcal{G}, \mathcal{H}} = \{ \mathbf{d} \in \mathbb{Z}^{|\mathcal{H}|} : d_i \geq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor, \text{ for } 0 \leq i \leq |\mathcal{H}|, \text{ and } \sum_{i=1}^{|\mathcal{H}|} d_i = |\mathcal{G}| \},$$

and $\mathbb{B}_{\mathcal{G}, \mathcal{H}}$ is a subset of $\mathbb{D}_{\mathcal{G}, \mathcal{H}}$.

Proof. Firstly, for $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|}) \in \mathbb{Z}^{|\mathcal{H}|}$ with $\sum_{i=1}^{|\mathcal{H}|} d_i = |\mathcal{G}|$, we show that if $\min(\mathbf{d}) \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$, then $\max(\mathbf{d}) > \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$, or $\max(\mathbf{d}) \geq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + 1$, where $\min(\mathbf{d})$ ($\max(\mathbf{d})$) denotes the minimum (maximum) component of vector $\mathbf{d}$. Otherwise, we have

$$\sum_{i=1}^{|\mathcal{H}|} d_i < \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + (|\mathcal{H}| - 1) \cdot \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor = |\mathcal{H}| \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor < |\mathcal{H}| \frac{|\mathcal{G}|}{|\mathcal{H}|} = |\mathcal{G}|.$$

For a given $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|}) \in \mathbb{Z}^{|\mathcal{H}|}$ with $\sum_{i=1}^{|\mathcal{H}|} d_i = |\mathcal{G}|$ and $\min(\mathbf{d}) \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$, we can transform it into another $\mathbf{d}' = (d'_1, \dots, d'_{|\mathcal{H}|}) \in \mathbb{Z}^{|\mathcal{H}|}$ with $\sum_{i=1}^{|\mathcal{H}|} d'_i = |\mathcal{G}|$ and $\min(\mathbf{d}') \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ by increasing the smallest entry of $\mathbf{d}$ by 1 and decreasing the maximal entry of $\mathbf{d}$ by 1. Next, we show that

$$\sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \geq \sum_{i=1}^{|\mathcal{H}|} \left(d'_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2. \quad (8)$$

Let $\min(\mathbf{d}) = \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor - \nabla$ and $\max(\mathbf{d}) = \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + \Delta$ with $\nabla \geq 0$ and $\Delta \geq 1$, and $u = \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor - \frac{|\mathcal{G}|}{|\mathcal{H}|}$. Then,

$$\begin{aligned} & \left(\min(\mathbf{d}) - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + \left(\max(\mathbf{d}) - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ & - \left(\min(\mathbf{d}') - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 - \left(\max(\mathbf{d}') - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ & = (u - \nabla)^2 + (u + \Delta)^2 - (u - \nabla + 1)^2 - (u + \Delta - 1)^2 \\ & = 2(\nabla + \Delta) - 2 \geq 0. \end{aligned}$$

Therefore, we can apply the above transformation for $\mathbf{d} \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}$ with $\min(\mathbf{d}) < \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ repeatedly, until $\min(\mathbf{d}) \geq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$.

Then, when $|\mathcal{H}| \nmid |\mathcal{G}|$ and $\rho \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$, we can get $\xi_{\mathcal{G}, \mathcal{H}}$.

Lemma 15. *For the two finite Abelian groups $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$, when $|\mathcal{H}| \nmid |\mathcal{G}|$ and $\rho \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$, we have*

$$\zeta_{\mathcal{G}, \mathcal{H}} = r \cdot \left(\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + 1 - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + (|\mathcal{H}| - r) \cdot \left(\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2.$$

Accordingly,

$$\xi_{\mathcal{G}, \mathcal{H}} = \left(r \cdot \left(\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + 1 \right)^2 + (|\mathcal{H}| - r) \cdot \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor^2 \right),$$

where $r = |\mathcal{G}| - |\mathcal{H}| \cdot \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ and $r < |\mathcal{H}|$.

Proof. For a given $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|}) \in \mathbb{B}_{\mathcal{G}, \mathcal{H}}$, we assume that there are k entries of $\mathbf{d}$ strictly larger than $\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$. Without loss of generality, we may assume that the k entries are $d_1, \dots, d_k$, and thus $d_{k+1} = \dots = d_{|\mathcal{H}|} = \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$. Let

$$d_1 = \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + \epsilon_1, \dots, d_k = \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + \epsilon_k$$

with $\epsilon_1, \dots, \epsilon_k \geq 1$, and $r = \sum_{j=1}^k \epsilon_j$. Then

$$|\mathcal{G}| = d_1 + \dots + d_k + d_{k+1} + \dots + d_{|\mathcal{H}|} = r + |\mathcal{H}| \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor. \quad (9)$$

Therefore, $r = |\mathcal{G}| - |\mathcal{H}| \cdot \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$. Note that $r < |\mathcal{H}|$, otherwise $r + |\mathcal{H}| \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor \geq |\mathcal{H}|(1 + \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor) > |\mathcal{H}| \frac{|\mathcal{G}|}{|\mathcal{H}|} = |\mathcal{G}|$.

Since $|\mathcal{H}| - k > r - k = \sum_{j=1}^k (\epsilon_j - 1)$, for each $j \in \{1, \dots, k\}$, we can select a new set of $\epsilon_j - 1$ entries of $\mathbf{d}$ (denoted as $d_{j_1}, \dots, d_{j_{\epsilon_j-1}}$) whose values are all $\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$. Based on the vector $\mathbf{d}$, we construct a modified vector $\mathbf{d}'$ by updating the entries corresponding to $d_{j_1}, \dots, d_{j_{\epsilon_j-1}}$ and d_j for each $j \in \{1, \dots, k\}$. Specifically, these entries are set to $\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + 1$. For the new vector $\mathbf{d}'$, it is easy to see that $\mathbf{d}' \in \mathbb{B}_{\mathcal{G}, \mathcal{H}}$. The vector $\mathbf{d}'$ is such that r components are equal to $\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + 1$ and the remaining $|\mathcal{H}| - r$ components are equal to $\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$.

Let $m = \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + 1 - \frac{|\mathcal{G}|}{|\mathcal{H}|}$, for any positive integer a , the following inequality holds

$$(m + a - 1)^2 + (a - 1)(m - 1)^2 \geq am^2.$$

And it is easy to verify that the inequality is equivalent to the inequality $a^2 \geq a$ where $a \geq 1$.

Because of the above inequality, for each $j \in \{1, \dots, k\}$, we can see that

$$\begin{aligned} & \left(d_j - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + \sum_{i=1}^{\epsilon_j-1} \left(d_{j_i} - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ &= (m + \epsilon_j - 1)^2 + (\epsilon_j - 1)(m - 1)^2 \\ &\geq \epsilon_j m^2 = \left(d'_j - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + \sum_{i=1}^{\epsilon_j-1} \left(d'_{j_i} - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2, \end{aligned}$$

where $m = \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + 1 - \frac{|\mathcal{G}|}{|\mathcal{H}|}$.

Thus, for all $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|}) \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}$ such that $d_i \geq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ where $1 \leq i \leq |\mathcal{H}|$, we have

$$\begin{aligned} & \sum_{j=1}^{|\mathcal{H}|} \left(d_j - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ &\geq \sum_{j=1}^{|\mathcal{H}|} \left(d'_j - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ &= r \cdot \left(\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + 1 - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + (|\mathcal{H}| - r) \cdot \left(\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2. \end{aligned}$$

In summary, we know that

$$\zeta_{\mathcal{G}, \mathcal{H}} = r \cdot \left(\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + 1 - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + (|\mathcal{H}| - r) \cdot \left(\left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2,$$

where $r = |\mathcal{G}| - |\mathcal{H}| \cdot \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$, $r < |\mathcal{H}|$.

Obviously, according to Lemma 12, if we know the value of $\zeta_{\mathcal{G}, \mathcal{H}}$, then we can get $\xi_{\mathcal{G}, \mathcal{H}}$ immediately.

Cases: $\rho > \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ and ρ divide $|\mathcal{G}|$

Lemma 16. For the two finite Abelian groups $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$, when $\rho > \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ and ρ divides $|\mathcal{G}|$, we have

$$\zeta_{\mathcal{G}, \mathcal{H}} = |\mathcal{G}| \rho - \frac{|\mathcal{G}|^2}{|\mathcal{H}|}.$$

Accordingly,

$$\xi_{\mathcal{G}, \mathcal{H}} = |\mathcal{G}| \rho.$$

Proof. For any $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|}) \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}$, for $1 \leq i \leq |\mathcal{H}|$, due to $d_i \geq \rho$ or $d_i = 0$, we have

$$\left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|}\right)^2 \geq \left(\rho - 2\frac{|\mathcal{G}|}{|\mathcal{H}|}\right) \cdot d_i + \frac{|\mathcal{G}|^2}{|\mathcal{H}|^2},$$

and when $d_i = 0$ or $d_i = \rho$, the equality holds.

For $\mathbf{d}' \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}$, we suppose that $d'_i = \rho$ for $1 \leq i \leq \frac{|\mathcal{G}|}{\rho}$ and $d'_i = 0$ for $\frac{|\mathcal{G}|}{\rho} + 1 \leq i \leq |\mathcal{H}|$. Thus, according to the above inequality for any $\mathbf{d} \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}$, we have

$$\begin{aligned} \sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|}\right)^2 &\geq \sum_{i=1}^{|\mathcal{H}|} \left(\rho - 2\frac{|\mathcal{G}|}{|\mathcal{H}|}\right) \cdot d'_i + \sum_{i=1}^{|\mathcal{H}|} \frac{|\mathcal{G}|^2}{|\mathcal{H}|^2} \\ &= \sum_{i=1}^{|\mathcal{G}|/\rho} \left(\rho^2 - \frac{|\mathcal{G}|}{|\mathcal{H}|} \rho\right) + \frac{|\mathcal{G}|^2}{|\mathcal{H}|} \\ &= |\mathcal{G}| \rho - \frac{|\mathcal{G}|^2}{|\mathcal{H}|}. \end{aligned}$$

In summary, we know that

$$\zeta_{\mathcal{G}, \mathcal{H}} = |\mathcal{G}| \rho - \frac{|\mathcal{G}|^2}{|\mathcal{H}|}.$$

Obviously, according to Lemma 12, if we know the value of $\zeta_{\mathcal{G}, \mathcal{H}}$, then we can get $\xi_{\mathcal{G}, \mathcal{H}}$ immediately.

Cases: $\rho > \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ and $\rho \nmid |\mathcal{G}|$

Lemma 17. Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups with $\rho > \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ and $\rho \nmid |\mathcal{G}|$. Then

$$\zeta_{\mathcal{G}, \mathcal{H}} = \min_{\mathbf{d} \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}} \sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|}\right)^2 = \min_{\mathbf{d} \in \mathbb{B}_{\mathcal{G}, \mathcal{H}}} \sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|}\right)^2,$$

where $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|})$, $m \cdot \rho = |\mathcal{G}| - n$,

$$\mathbb{D}_{\mathcal{G}, \mathcal{H}} = \{\mathbf{d} \in \mathbb{Z}^{|\mathcal{H}|} : d_i = 0 \text{ or } d_i \geq \rho, \text{ for } 0 \leq i \leq |\mathcal{H}|, \text{ and } \sum_{i=1}^{|\mathcal{H}|} d_i = |\mathcal{G}|\},$$

$\mathbb{B}_{\mathcal{G}, \mathcal{H}} = \{\mathbf{d} \in \mathbb{D}_{\mathcal{G}, \mathcal{H}} : \text{for } 0 \leq i \leq |\mathcal{H}|, \text{ the number of } d_i \text{ such that } d_i > 0 \text{ is } m\}.$
and $\mathbb{B}_{\mathcal{G}, \mathcal{H}}$ is a subset of $\mathbb{D}_{\mathcal{G}, \mathcal{H}}$.

Proof. Suppose that $|\mathcal{G}| = m \cdot \rho + n$, $n < \rho$. Then, we can see that the number of nonzero components is at most m . Otherwise, it will be in contradiction to $n < \rho$.

And we define the set $\mathbb{D}'_{\mathcal{G}, \mathcal{H}}$ as

$$\mathbb{D}'_{\mathcal{G}, \mathcal{H}} = \{\mathbf{h} \in \mathbb{Z}^{|\mathcal{H}|} : h_i = 0 \text{ or } h_i \geq \rho, \text{ for } 0 \leq i \leq |\mathcal{H}|, \\ \text{and } \sum_{i=1}^{|\mathcal{H}|} h_i = |\mathcal{G}| - n\},$$

where $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|})$, $n = |\mathcal{G}| - m \cdot \rho$.

For $\mathbf{h}' = (h'_1, \dots, h'_{|\mathcal{H}|}) \in \mathbb{D}'_{\mathcal{G}, \mathcal{H}}$, we suppose that $h'_i = \rho$, $1 \leq i \leq m$ and $h'_i = 0$, $m+1 \leq i \leq |\mathcal{H}|$. Then, like Lemma 16, using same technique, we get

$$\sum_{i=1}^{|\mathcal{H}|} \left(h'_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 = \min_{\mathbf{h} \in \mathbb{D}'_{\mathcal{G}, \mathcal{H}}} \sum_{i=1}^{|\mathcal{H}|} \left(h_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2. \quad (10)$$

Next, for any $\mathbf{d} = (d_1, \dots, d_H) \in \mathbb{D}_{\mathcal{G}, \mathcal{H}}$, we assume that there are p ($p \leq m < |\mathcal{H}|$) nonzero components. For convenience, we denote it as the first p components. Then, we define $\mathbf{d}' \in \mathbb{D}'_{\mathcal{G}, \mathcal{H}}$ from $\mathbf{d}$, where $d'_i = 0$ for $p+1 \leq i \leq |\mathcal{H}|$ and $d'_i = d_i - r_i$, $d_i \geq \rho$, $0 \leq r_i \leq d_i - \rho$ for $1 \leq i \leq p$. In addition, $\sum_{i=0}^p r_i = n$. And we also define $\mathbf{d}^b = (d_1^b, \dots, d_{|\mathcal{H}|}^b) \in \mathbb{B}_{\mathcal{G}, \mathcal{H}}$, where $d_i^b = h'_i + r_i = \rho + r_i$ for $1 \leq i \leq p$ and $d_i^b = h'_i$ for $p+1 \leq i \leq |\mathcal{H}|$.

Then, for $1 \leq i \leq p$, due to

$$(2d'_i + r_i - 2\frac{|\mathcal{G}|}{|\mathcal{H}|}) r_i \geq (2\rho + r_i - 2\frac{|\mathcal{G}|}{|\mathcal{H}|}) r_i,$$

we can see that

$$\begin{aligned} & \left(d'_i + r_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 - \left(d'_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ & \geq \left(\rho + r_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 - \left(\rho - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2. \end{aligned} \quad (11)$$

Finally, according to Equation (10) and Inequality (11), we have

$$\begin{aligned} & \left(\sum_{i=0}^p \left(d'_i + r_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 - \left(d'_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \right) + \sum_{i=0}^{|\mathcal{H}|} \left(d'_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ & = \sum_{i=0}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ & \geq \left(\sum_{i=0}^p \left(\rho + r_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 - \left(\rho - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \right) + \sum_{i=0}^{|\mathcal{H}|} \left(h'_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ & = \sum_{i=0}^{|\mathcal{H}|} \left(d_i^b - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2. \end{aligned}$$

Then, when $\rho > \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ and $\rho \nmid |\mathcal{G}|$, we can get $\xi_{\mathcal{G}, \mathcal{H}}$.

Lemma 18. *For the two finite Abelian groups $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$, when $\rho > \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ and $\rho \nmid |\mathcal{G}|$, we have*

$$\zeta_{\mathcal{G}, \mathcal{H}} = \left((m - h) \cdot \left(c - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + h \cdot \left(c + 1 - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \right).$$

Accordingly,

$$\xi_{\mathcal{G}, \mathcal{H}} = (m - h) \cdot c^2 + h \cdot (c + 1)^2$$

where $m \cdot \rho = |\mathcal{G}| - n$, $n < \rho$ and $h = |\mathcal{G}| - c \cdot m$, $h < m$.

Proof. For any $\mathbf{d} = (d_1, \dots, d_{|\mathcal{H}|}) \in \mathbb{B}_{\mathcal{G}, \mathcal{H}}$, for convenience, we can let $d_i = \rho + r_i$ for $1 \leq i \leq m$ and $d_i = 0$ for $m + 1 \leq i \leq |\mathcal{H}|$. In addition, $\sum_{i=1}^{|\mathcal{H}|} r_i = n$, where $n = |\mathcal{G}| - m \cdot \rho$.

Let $w = \rho - \frac{|\mathcal{G}|}{|\mathcal{H}|}$, then we can see that, for any $z, j \in \{1, \dots, m\}$ such that $r_z > r_j$, we have

$$(w + r_z)^2 + (w + r_j)^2 \geq (w + r_z - 1)^2 + (w + r_j + 1)^2,$$

and the equation holds if and only if $r_z - r_j = 1$.

For $\mathbf{d}' = (d'_1, \dots, d'_{|\mathcal{H}|}) \in \mathbb{B}_{\mathcal{G}, \mathcal{H}}$, we let $d'_i = d_i$, $i \in \{1, \dots, m\}$ except for z, j and $r'_z = r_z - 1$, $r'_j = r_j + 1$. Then, from the inequality above, we have

$$\sum_{i=0}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \geq \sum_{i=0}^{|\mathcal{H}|} \left(d'_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2,$$

and the equation holds if and only if $r_z - r_j = 1$.

Thus, for $\mathbf{d}' = (d'_1, \dots, d'_{|\mathcal{H}|}) \in \mathbb{B}_{\mathcal{G}, \mathcal{H}}$ such that $d'_z, d'_j \geq \rho$ and $|d'_z - d'_j| \leq 1$ holds for any $z, j \in \{1, \dots, |\mathcal{H}|\}$, we have

$$\begin{aligned} & \sum_{i=1}^{|\mathcal{H}|} \left(d'_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \\ &= \left((m - h) \cdot \left(c - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + h \cdot \left(c + 1 - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \right) \\ &= \min_{\mathbf{d} \in \mathbb{B}_{\mathcal{G}, \mathcal{H}}} \sum_{i=1}^{|\mathcal{H}|} \left(d_i - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2, \end{aligned}$$

where $m \cdot \rho = |\mathcal{G}| - n$, $n < \rho$ and $h = |\mathcal{G}| - c \cdot m$, $h < m$.

According to Lemma 17,

$$\zeta_{\mathcal{G}, \mathcal{H}} = \left((m - h) \cdot \left(c - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 + h \cdot \left(c + 1 - \frac{|\mathcal{G}|}{|\mathcal{H}|} \right)^2 \right),$$

where $m \cdot \rho = |\mathcal{G}| - n$, $n < \rho$ and $h = |\mathcal{G}| - c \cdot m$, $h < m$.

By gathering Lemma 13, Lemma 15, Lemma 16, Lemma 18, according to $|\mathcal{G}|$, $|\mathcal{H}|$ and $\rho_{\mathcal{G},\mathcal{H}}^{\mathcal{F}}$, we get the value of $\xi_{\mathcal{G},\mathcal{H}}$.

Theorem 3. *For the two finite Abelian groups $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$, the value of $\xi_{\mathcal{G},\mathcal{H}}$ is shown in Table 1.*

Table 1: Summary of the value of $\xi_{\mathcal{G},\mathcal{H}}$		
Cases	$\xi_{\mathcal{G},\mathcal{H}}$	
$\rho \leq \left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor$	$ \mathcal{H} \text{ divide } \mathcal{G} $	$\frac{ \mathcal{G} ^2}{ \mathcal{H} }$
	$ \mathcal{H} \nmid \mathcal{G} $	$(\mathcal{H} - r) \cdot \left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor^2 + r \cdot \left(\left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor + 1 \right)^2, \quad r = \mathcal{G} - \mathcal{H} \cdot \left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor, r < \mathcal{H} $
$\rho > \left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor$	$\rho \text{ divide } \mathcal{G} $	$ \mathcal{G} \cdot \rho$
	$\rho \nmid \mathcal{G} $	$(m - h) \cdot c^2 + h \cdot (c + 1)^2, \quad \begin{cases} m \cdot \rho = \mathcal{G} - n, n < \rho \\ h = \mathcal{G} - c \cdot m, h < m \end{cases}$

4.2 The tighter bound of λ_F

Then, we will derive a tighter bound of λ_F . According to the Parseval equation, we can derive a relationship between λ_F and $\sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}, \chi \neq \mathbf{1}_{\mathcal{H}}} \|\mathcal{C}_{\psi,\chi}^F\|^4$.

Lemma 19. *It holds that*

$$(\lambda_F)^2 \cdot (|\mathcal{H}| - 1) \geq \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}, \chi \neq \mathbf{1}_{\mathcal{H}}} \|\mathcal{C}_{\psi,\chi}^F\|^4.$$

The equality holds if and only if the equation holds $\|\mathcal{C}_{\psi,\chi}^F\| = \lambda_F$ or $\|\mathcal{C}_{\psi,\chi}^F\| = 0$, for all $\psi \in \hat{\mathcal{G}}$, $\chi \in \hat{\mathcal{H}}$, and $\chi \neq \mathbf{1}_{\mathcal{H}}$.

Proof. For any $\psi \in \hat{\mathcal{G}}, \chi \in \hat{\mathcal{H}}$ and $\chi \neq \mathbf{1}_{\mathcal{H}}$, we have

$$(\lambda_F)^2 \cdot \|\mathcal{C}_{\psi,\chi}^F\|^2 \geq \|\mathcal{C}_{\psi,\chi}^F\|^4,$$

and the equation holds if and only if $\|\mathcal{C}_{\psi,\chi}^F\| = \lambda_F$ or $\|\mathcal{C}_{\psi,\chi}^F\| = 0$. Thus,

$$(\lambda_F)^2 \cdot \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}, \chi \neq \mathbf{1}_{\mathcal{H}}} \|\mathcal{C}_{\psi,\chi}^F\|^2 \geq \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}, \chi \neq \mathbf{1}_{\mathcal{H}}} \|\mathcal{C}_{\psi,\chi}^F\|^4.$$

Combined with Theorem 2, we get the inequality.

Together with Theorem 1, we get the following result.

Lemma 20. *Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. For a set $\mathcal{F}$ of functions from a finite Abelian group $(\mathcal{G}, +)$ to another finite Abelian group $(\mathcal{H}, +)$, we have*

$$\sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \mathcal{H}, \chi \neq 1_{\mathcal{H}}} \|\mathcal{C}_{\psi, \chi}^F\|^4 \geq \frac{|\mathcal{H}|(|\mathcal{G}| - 1)}{|\mathcal{G}|^3} \cdot \xi_{\mathcal{G}, \mathcal{H}} + \frac{|\mathcal{H}|}{|\mathcal{G}|} - 1,$$

where the value of $\xi_{\mathcal{G}, \mathcal{H}}$ can be obtained in Theorem 3.

Proof. According to the Theorem 1 and Equation 6, we have

$$\begin{aligned} \sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \mathcal{H}, \chi \neq 1} \|\mathcal{C}_{\psi, \chi}^F\|^4 &= \frac{|\mathcal{H}|}{|\mathcal{G}|^3} \sum_{\alpha \in \mathcal{G}, \alpha \neq 0} \sum_{\beta \in \mathcal{H}} \delta_F(\alpha, \beta)^2 + \frac{|\mathcal{H}|}{|\mathcal{G}|} - 1 \\ &\geq \frac{|\mathcal{H}|}{|\mathcal{G}|^3} \sum_{\alpha \in \mathcal{G}, \alpha \neq 0} \xi_{\mathcal{G}, \mathcal{H}} + \frac{|\mathcal{H}|}{|\mathcal{G}|} - 1 \\ &= \frac{|\mathcal{H}|(|\mathcal{G}| - 1)}{|\mathcal{G}|^3} \cdot \xi_{\mathcal{G}, \mathcal{H}} + \frac{|\mathcal{H}|}{|\mathcal{G}|} - 1. \end{aligned}$$

Remark 2. It is worth noting that when $\mathcal{G} = \mathbb{F}_2^m$, $\mathcal{H} = \mathbb{F}_2^n$, the lemma is consistent with Eq.(15) in [CD07].

Then, combined with Lemma 19, Lemma 20, Theorem 3, according to $|\mathcal{G}|$, $|\mathcal{H}|$ and $\rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}}$, we get the lower bound on λ_F and complete the generalization of Sidelnikov-Chabaud-Vaudenay's bound from $\mathbb{F}_2^n$ to finite Abelian groups.

Theorem 4. *Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. For a set $\mathcal{F}$ of functions from a finite Abelian group $(\mathcal{G}, +)$ to another finite Abelian group $(\mathcal{H}, +)$, we have*

$$\lambda_F \geq \sqrt{\frac{1}{|\mathcal{H}| - 1} \left(\frac{|\mathcal{H}|(|\mathcal{G}| - 1)}{|\mathcal{G}|^3} \cdot \xi_{\mathcal{G}, \mathcal{H}} + \frac{|\mathcal{H}|}{|\mathcal{G}|} - 1 \right)},$$

where the value of $\xi_{\mathcal{G}, \mathcal{H}}$ can be obtained in Theorem 3, and the low bound of λ_F is summarized in Table 2.

Remark 3. For the condition $|\mathcal{G}| \mid |\mathcal{H}|$ and $\rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}} \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$, [Pot04] has proved that the optimal function against differential cryptanalysis satisfies $\lambda_F = (\sqrt{|\mathcal{G}|})^{-1}$.

Remark 4. If we consider the set $\mathcal{F}$ of functions from finite Abelian group $(\mathcal{G}, +)$ to another finite Abelian group $(\mathcal{H}, +)$ such that $\rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}} = |\mathcal{G}|$ (The set contains all “affine” functions from a finite Abelian group $(\mathcal{G}, +)$ to another finite Abelian group $(\mathcal{H}, +)$), then we have $\lambda_F \geq 1$ from Theorem 4.

Remark 5. For the function set $\mathcal{F}$ such that $\rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}} > \frac{|\mathcal{G}|}{2}$, according to Theorem 4, we can get $\lambda_F \geq 1$.

Table 2: Summary of the low bound of λ_F

Cases	The low bound of λ_F	
$\rho_{\mathcal{G}, \mathcal{H}} \leq \left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor$	$ \mathcal{H} \text{ divide } \mathcal{G} $	$ \mathcal{G} ^{-\frac{1}{2}}$
	$ \mathcal{H} \nmid \mathcal{G} $	$\sqrt{\frac{ \mathcal{H} (\mathcal{G} -1)}{ \mathcal{G} ^3(\mathcal{H} -1)}} \left(\mathcal{H} \cdot \left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor^2 + 2r \cdot \left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor + r \right) + \frac{ \mathcal{H} - \mathcal{G} }{ \mathcal{G} (\mathcal{H} -1)},$ $r = \mathcal{G} - \mathcal{H} \cdot \left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor, r < \mathcal{H} $
$\rho_{\mathcal{G}, \mathcal{H}} > \left\lfloor \frac{ \mathcal{G} }{ \mathcal{H} } \right\rfloor$	$\rho_{\mathcal{G}, \mathcal{H}} \text{ divide } \mathcal{G} $	$\sqrt{\frac{ \mathcal{H} (\mathcal{G} -1)}{ \mathcal{G} ^2(\mathcal{H} -1)}} \rho_{\mathcal{G}, \mathcal{H}} + \frac{ \mathcal{H} - \mathcal{G} }{ \mathcal{G} (\mathcal{H} -1)}$
	$\rho_{\mathcal{G}, \mathcal{H}} \nmid \mathcal{G} $	$\sqrt{\frac{ \mathcal{H} (\mathcal{G} -1)}{ \mathcal{G} ^3(\mathcal{H} -1)}} ((m-h) \cdot c^2 + h \cdot (c+1)^2) + \frac{ \mathcal{H} - \mathcal{G} }{ \mathcal{G} (\mathcal{H} -1)},$ $m \cdot \rho = \mathcal{G} - n, n < \rho, h = \mathcal{G} - c \cdot m, h < m$

Example 12. The nonlinear function $F(x) = x^s$ over the finite field $\mathbb{F}_p$ ($\gcd(s, p) = 1$) is the most common non-linear functions used in both the Feistel structure (GMiMC [AGP⁺19], MiMC-2n/n [AGR⁺16]) and SPN structure (MiMC-n/n [AGR⁺16], HadesMiMC [GLR⁺20]). To validate the correctness of Theorem 4, we compute the λ_F of $F(x) = x^3 \bmod p$ for certain prime numbers and compare it with the lower bound provided in Theorem 4. Additionally, we also use the function $F(x) = ((-1)^{x \bmod 2} \cdot x^2) \bmod p$, which was proposed in [GKRS22] (see the fifth row of Table 1 in [GKRS22]), to further verify the correctness of Theorem 4. As illustrated in Fig. 1, when both the input and output domains are set to $\mathbb{F}_p$, the values of λ_F for the aforementioned functions are not less than the lower bound specified in Theorem 4.

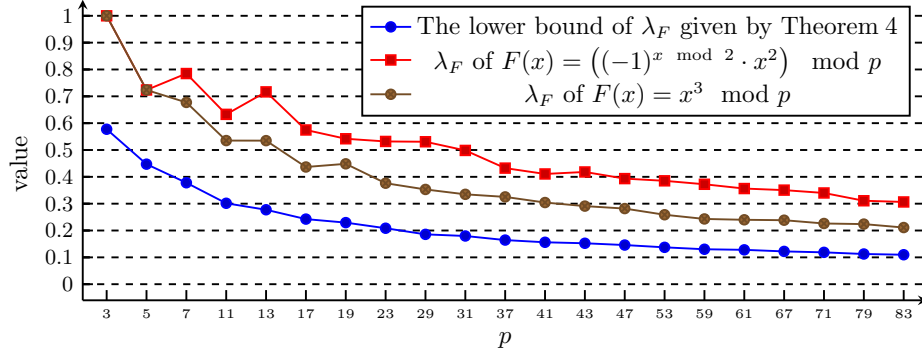


Fig. 1: The value of λ_F of $F(x) = x^3 \bmod p$, $F(x) = ((-1)^{x \bmod 2} \cdot x^2) \bmod p$ and the value of lower bound given in Theorem 4 under different p .

Moreover, when the input domain and output domain are set to $\mathbb{F}_q$ and $\mathbb{F}_q$ respectively, where $q > p$, the same result is demonstrated in Fig. 2. Similarly, for the case where $q < p$, we observe the same result, as shown in Fig. 3.

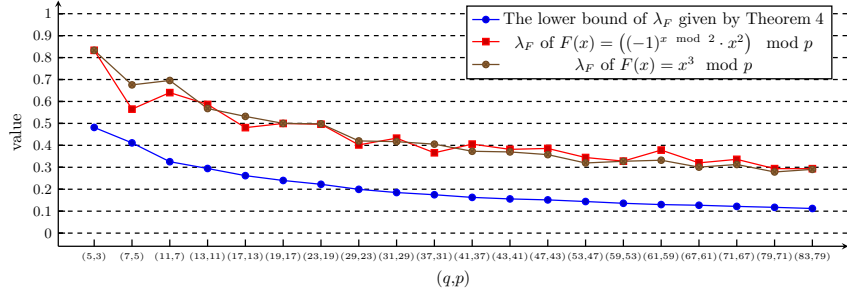


Fig. 2: The value of λ_F of $F(x) = x^3 \mod p$, $F(x) = ((-1)^{x \mod 2} \cdot x^2) \mod p$ and the value of lower bound given in Theorem 4 under different (p, q) , where $q > p$.

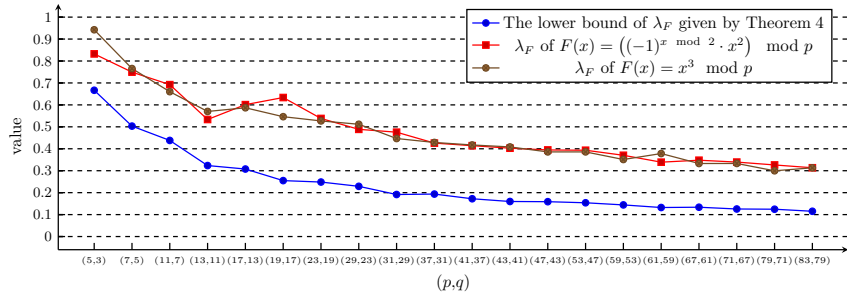


Fig. 3: The value of λ_F of $F(x) = x^3 \mod p$, $F(x) = ((-1)^{x \mod 2} \cdot x^2) \mod p$ and the value of lower bound given in Theorem 4 under different (p, q) , where $q < p$.

5 Differential-linear Approximations over Finite Abelian Groups

As the second application of the link between differential cryptanalysis and linear cryptanalysis, we can extend the closed formula of differential-linear approximations, which is the exact formula derived under the sole assumption of independence, from $\mathbb{F}_2^n$ to finite Abelian groups.

5.1 Background

Differential-linear approximations over finite fields of characteristic 2 Given a function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, the differential-linear approximation over finite fields of characteristic 2 with the input difference $\alpha \in \mathbb{F}_2^n$ and the output mask $\lambda \in \mathbb{F}_2^m$ is defined as

$$DL_{\alpha, \lambda}^F = \frac{1}{2^n} \sum_{x \in \mathbb{F}_2^n} (-1)^{\lambda \cdot (F(x) \oplus F(x \oplus \alpha))}.$$

For the differential-linear approximations over finite fields of characteristic 2, Blondeau et al. [BLN17] have proved the following theorem since Langford and Hellman [LH94] proposed the differential-linear cryptanalysis.

Theorem 5 ([BLN17], **Theorem 2**). *Let E be the cipher over $\mathbb{F}_2^n$ that can be divided into $E = E_1 \circ E_0$. If E_0 and E_1 are independent, the correlation of a differential-linear distinguisher with the input difference α and the output linear mask λ can be computed as*

$$DL_{\alpha,\lambda}^E = \sum_{v \in \mathbb{F}_2^n} DL_{\alpha,\gamma}^{E_0} (C_{\gamma,\lambda}^{E_1})^2, \quad (12)$$

for all $\alpha \neq 0$ and $\gamma \neq 0$.

The above theorem illustrates that the differential-linear approximations can be divided into two parts as shown in Fig. 4.

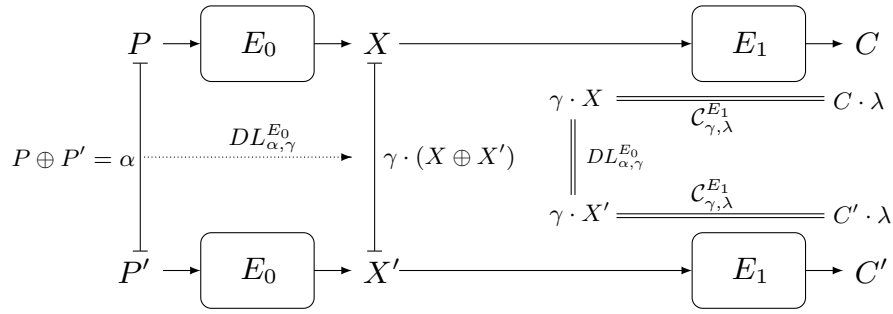


Fig. 4: A high-level illustration of a differential-linear approximation.

Later, Bar-On et al. [BDKW19] proposed the concept of DLCT, and based on Theorem 5 they gave the following result, which is shown in Fig 5.

Theorem 6 ([BDKW19], **Equation (5)**). *Let E be the cipher over $\mathbb{F}_p^n$ that can be divided into $E = E_1 \circ E_m \circ E_0$. If E_0 , E_m and E_1 are independent, the correlation of a differential-linear distinguisher with input difference α and output linear mask λ can be computed as*

$$DL_{\alpha,\lambda}^E = \sum_{\delta \in \mathbb{F}_2^n} \Pr[\alpha \xrightarrow{E_0} \delta] \sum_{\gamma \in \mathbb{F}_2^n} DL_{\delta,\gamma}^{E_m} (C_{\gamma,\lambda}^{E_1})^2, \quad (13)$$

for all $\delta \neq 0$ and $\gamma \neq 0$.

In addition, if we suppose that $DL_{\delta',\gamma'}^{E_m} = 0$ for all $\delta' \neq \delta$, $\gamma' \neq \gamma$, then we have

$$DL_{\alpha,\lambda}^E = \Pr[\alpha \xrightarrow{E_0} \delta] DL_{\delta,\gamma}^{E_m} (C_{\gamma,\lambda}^{E_1})^2. \quad (14)$$

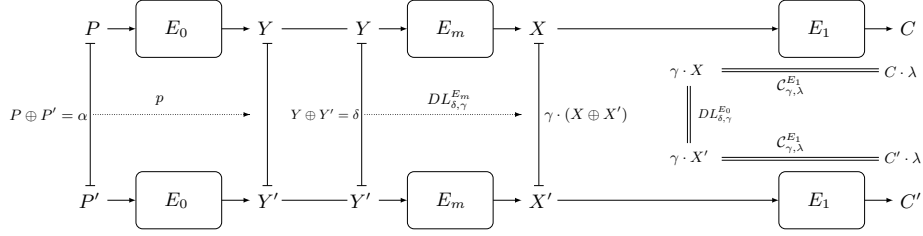


Fig. 5: A high-level illustration of a differential-linear approximation.

Differential-linear approximations over $\mathbb{F}_p$ Inspired by Equation (14) on the differential-linear approximation over $\mathbb{F}_p$, Xu et al. [XCWW23] proposed the following framework.

Let E be the cipher over $\mathbb{F}_p^n$ that can be divided into $E = E_1 \circ E_m \circ E_0$, and we consider these three parts separately as follows:

- For E_0 , it has a differential $\alpha \rightarrow \delta$, $\delta, \alpha \in \mathbb{F}_p^n$, with probability $\Pr[\alpha \xrightarrow{E_0} \delta]$.
- For E_1 , it has a linear approximations $\gamma \rightarrow \lambda$, $\lambda, \gamma \in \mathbb{F}_p^n$, with the correlation

$$C_{\gamma, \lambda}^{E_1} = \frac{1}{p^n} \sum_{x \in \mathbb{F}_p^n} e^{-\frac{2\pi i}{p}(\gamma \cdot x)} e^{\frac{2\pi i}{p}(\lambda \cdot E_1(x))},$$

where $\gamma \cdot x$ is the inner product over $\mathbb{F}_p^n$, and $e^{\frac{2\pi i}{p}(\gamma \cdot x)}$ is the group character $\chi(x)$ of $\mathbb{F}_p^n$.

-For E_m , it has differential-linear approximations $\delta \rightarrow \gamma$, $\gamma, \delta \in \mathbb{F}_p^n$, with the correlation

$$DL_{\delta, \gamma}^{E_m} = \frac{1}{p^n} \sum_{x \in \mathbb{F}_p^n} e^{\frac{2\pi i}{p}(\gamma \cdot (E_m(x) - E_m(x - \delta)))},$$

where $v \cdot x$ is the inner product over $\mathbb{F}_p^n$.

Beside this, under these three assumptions as below, we can get the correlation of differential-linear approximations $\alpha \rightarrow \gamma$ of E , which is illustrated in Theorem 7.

- E_0 , E_m and E_1 are independent.

-For all $\delta' \neq \delta$, $\gamma' \neq \gamma$, $DL_{\delta', \gamma'}^{E_m} = 0$.

-Let $X = E_m \circ E_0(P)$, $X' = E_m \circ E_0(P - \alpha)$, X and X' are independent of each other.

Theorem 7 ([XCWW23], Theorem 3). *For the block cipher $E : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$, $E = E_1 \circ E_m \circ E_0$, the correlation of a differential-linear distinguisher with the input difference $\alpha \in \mathbb{F}_p^n$ and the output linear mask $\lambda \in \mathbb{F}_p^n$ can be computed as*

$$DL_{\alpha, \gamma}^E = \Pr[\alpha \rightarrow \delta] DL_{\delta, \gamma}^{E_m} \|C_{\gamma, \lambda}^{E_1}\|^2. \quad (15)$$

Remark 6. Notably, we emphasize that the last two assumptions may not hold in practice, and therefore, a stronger version of Theorem 7 under the sole independent assumption is required.

Remark 7. For the differential-linear approximations over $\mathbb{F}_2^n$, it is easy to convert it into a special multidimensional difference of which the output difference set is the orthogonal space of the output mask λ . For this reason, Blondeau et al. [BLN17] conclude the closed formula for differential-linear approximations over $\mathbb{F}_2^n$ based on the link between the multidimensional differential cryptanalysis and multidimensional linear approximations [BN13]. However, for the differential-linear approximations over $\mathbb{F}_p^n$, it can not be converted into multidimensional difference directly. It seems that it is infeasible to drive the closed formula of differential-linear approximations over $\mathbb{F}_p^n$ by using the technique of Blondeau et al. [BLN17].

5.2 Differential-linear approximations over finite Abelian groups

In this part, we will generalize Theorem 5 from $\mathbb{F}_2^n$ to finite Abelian groups by a completely new technique. Our technique used in this part is not trivial. In brief, our technique is based on the link between the differential cryptanalysis and linear approximations. The differential-linear approximations over finite Abelian groups can be generalized in the following.

Definition 13. Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. The correlation of the differential-linear approximation of F with the input difference $\alpha \in \mathcal{G}$ and the output mask $\chi \in \hat{\mathcal{H}}$ is defined as

$$DL_{\alpha, \chi}^F = \frac{1}{|\mathcal{G}|} \sum_{x \in \mathcal{G}} \chi(F(x) - F(x - \alpha)).$$

Based on the link between the differential cryptanalysis and linear approximations, for the differential-linear approximations over finite Abelian groups, we have the following lemma. The lemma gives the link between the differential-linear approximations and linear approximations over finite Abelian groups.

Lemma 21. Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. We have

$$DL_{\alpha, \chi}^F = \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \mathcal{C}_{\psi, \bar{\chi}}^F \overline{\mathcal{C}_{\psi, \bar{\chi}}^F} = \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \|\mathcal{C}_{\bar{\psi}, \chi}^F\|^2.$$

Proof. Firstly, we have

$$\begin{aligned} DL_{\alpha, \chi}^F &= \frac{1}{|\mathcal{G}|} \sum_{\beta \in \mathcal{H}} \chi(\beta) \sum_{x \in \mathcal{G}} \Delta_{\beta, F(x) - F(x - \alpha)} \\ &= \frac{1}{|\mathcal{G}|} \sum_{\beta \in \mathcal{H}} \chi(\beta) \delta_F(\alpha, \beta). \end{aligned}$$

Applying the Lemma 11, we have

$$\begin{aligned}
DL_{\alpha,\chi}^F &= \frac{1}{|\mathcal{G}|} \sum_{\beta \in \mathcal{H}} \chi(\beta) \frac{|\mathcal{G}|}{|\mathcal{H}|} \sum_{\rho \in \hat{\mathcal{H}}, \psi \in \hat{\mathcal{G}}} \bar{\psi}(g) \rho(h) \mathcal{C}_{\psi,\rho}^F \overline{\mathcal{C}_{\psi,\rho}^F} \\
&= \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \sum_{\rho \in \hat{\mathcal{H}}} \frac{1}{|\mathcal{H}|} \sum_{\beta \in \mathcal{H}} \rho(\beta) \chi(\beta) \mathcal{C}_{\psi,\rho}^F \overline{\mathcal{C}_{\psi,\rho}^F} \\
&= \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \sum_{\rho \in \hat{\mathcal{H}}} \mathcal{C}_{\psi,\rho}^F \overline{\mathcal{C}_{\psi,\rho}^F} \frac{1}{|\mathcal{H}|} \sum_{\beta \in \mathcal{H}} \rho(\beta) \chi(\beta) \\
&= \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \sum_{\rho \in \hat{\mathcal{H}}} \mathcal{C}_{\psi,\rho}^F \overline{\mathcal{C}_{\psi,\rho}^F} \Delta_{\rho,\bar{\chi}} \quad (\text{Lemma 4}) \\
&= \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \mathcal{C}_{\psi,\bar{\chi}}^F \overline{\mathcal{C}_{\psi,\bar{\chi}}^F} \\
&= \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \|\mathcal{C}_{\psi,\chi}^F\|^2.
\end{aligned}$$

In addition, based on Lemma 11, for differentially independent functions, we can get the further link between the differential cryptanalysis and linear approximations.

Lemma 22. *Let $(\mathcal{G}, +)$, $(\mathcal{K}, +)$ and $(\mathcal{H}, +)$ be three finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function such that $F = F_2 \circ F_1$, where $F_1 : \mathcal{G} \rightarrow \mathcal{K}$ and $F_2 : \mathcal{K} \rightarrow \mathcal{H}$ are differentially independent, i.e., for all $\alpha \in \mathcal{G}$, $\delta \in \mathcal{K}$, $\beta \in \mathcal{H}$, we have $\Pr[\alpha \xrightarrow{F_1} \delta \xrightarrow{F_2} \beta] = \Pr[\alpha \xrightarrow{F_1} \delta] \Pr[\delta \xrightarrow{F_2} \beta]$. Then, we have for all $\chi \in \hat{\mathcal{H}}$ and $\psi \in \hat{\mathcal{G}}$,*

$$\|\mathcal{C}_{\psi,\chi}^F\|^2 = \sum_{\rho \in \hat{\mathcal{K}}} \|\mathcal{C}_{\psi,\rho}^{F_1}\|^2 \|\mathcal{C}_{\rho,\chi}^{F_2}\|^2.$$

Proof. According to the definition of differential independent, we have

$$\Pr[\alpha \xrightarrow{F} \beta] = \sum_{\delta \in \mathcal{K}} \Pr[\alpha \xrightarrow{F_1} \delta \xrightarrow{F_2} \beta] = \sum_{\delta \in \mathcal{K}} \Pr[\alpha \xrightarrow{F_1} \delta] \Pr[\delta \xrightarrow{F_2} \beta].$$

Then, applying Lemma 11, we have

$$\begin{aligned}
\|\mathcal{C}_{\psi,\chi}^F\|^2 &= \frac{1}{|\mathcal{G}|} \sum_{\alpha \in \mathcal{G}, \beta \in \mathcal{H}} \psi(\alpha) \bar{\chi}(\beta) \Pr[\alpha \xrightarrow{F} \beta] \\
&= \frac{1}{|\mathcal{G}|} \sum_{\alpha \in \mathcal{G}, \beta \in \mathcal{H}, \delta \in \mathcal{K}} \psi(g) \bar{\chi}(h) \Pr[\alpha \xrightarrow{F_1} \delta] \Pr[\delta \xrightarrow{F_2} \beta].
\end{aligned}$$

Furthermore, applying Lemma 11 again, we have

$$\begin{aligned}
& \frac{1}{|\mathcal{G}|} \sum_{\alpha \in \mathcal{G}, \beta \in \mathcal{H}, \delta \in \mathcal{K}} \psi(\alpha) \bar{\chi}(\beta) \Pr[\alpha \xrightarrow{F_1} \delta] \Pr[\delta \xrightarrow{F_2} \beta] \\
&= \frac{1}{|\mathcal{H}||\mathcal{K}||\mathcal{G}|} \sum_{\alpha \in \mathcal{G}, \beta \in \mathcal{H}, \delta \in \mathcal{K}} \psi(\alpha) \bar{\chi}(\beta) \sum_{\rho_1 \in \hat{\mathcal{K}}, \psi_1 \in \hat{\mathcal{G}}} \bar{\psi}_1(\alpha) \rho_1(\delta) \|\mathcal{C}_{\psi_1, \rho_1}^{F_1}\|^2 \\
&\quad \sum_{\chi_1 \in \hat{\mathcal{H}}, \rho_2 \in \hat{\mathcal{K}}} \bar{\rho}_2(\delta) \chi_1(\beta) \|\mathcal{C}_{\rho_2, \chi_1}^{F_2}\|^2 \\
&= \sum_{\rho_2, \rho_1 \in \hat{\mathcal{K}}} \sum_{\chi_1 \in \hat{\mathcal{H}}} \sum_{\psi_1 \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi_1, \rho_1}^{F_1}\|^2 \|\mathcal{C}_{\rho_2, \chi_1}^{F_2}\|^2 \left(\sum_{\delta \in \mathcal{K}} \frac{1}{|\mathcal{K}|} \bar{\rho}_2(\delta) \rho_1(\delta) \right) \\
&\quad \left(\sum_{\alpha \in \mathcal{G}} \frac{1}{|\mathcal{G}|} \bar{\psi}_1(\alpha) \psi(\alpha) \right) \left(\frac{1}{|\mathcal{H}|} \sum_{\beta \in \mathcal{H}} \chi_1(\beta) \bar{\chi}(\beta) \right) \\
&= \sum_{\rho_2, \rho_1 \in \hat{\mathcal{K}}} \sum_{\chi_1 \in \hat{\mathcal{H}}} \sum_{\psi_1 \in \hat{\mathcal{G}}} \|\mathcal{C}_{\psi_1, \rho_1}^{F_1}\|^2 \|\mathcal{C}_{\rho_2, \chi_1}^{F_2}\|^2 \Delta_{\rho_2, \rho_1} \Delta_{\psi, \psi_1} \Delta_{\chi, \chi_1} \quad (\text{Lemma 4}) \\
&= \sum_{\rho \in \hat{\mathcal{K}}} \|\mathcal{C}_{\psi, \rho}^{F_1}\|^2 \|\mathcal{C}_{\rho, \chi}^{F_2}\|^2.
\end{aligned}$$

Thus, we conclude that

$$\|\mathcal{C}_{\psi, \chi}^F\|^2 = \sum_{\rho \in \hat{\mathcal{K}}} \|\mathcal{C}_{\psi, \rho}^{F_1}\|^2 \|\mathcal{C}_{\rho, \chi}^{F_2}\|^2$$

holds.

Based on Lemma 21 and Lemma 22, we can complete the generalization of Theorem 5 in [BLN17] from $\mathbb{F}_2^n$ to finite Abelian groups.

Theorem 8. *Let $(\mathcal{G}, +)$, $(\mathcal{K}, +)$ and $(\mathcal{H}, +)$ be three finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function such that $F = F_2 \circ F_1$, where $F_1 : \mathcal{G} \rightarrow \mathcal{K}$ and $F_2 : \mathcal{K} \rightarrow \mathcal{H}$ are differentially independent. Then, for the correlation of the differential-linear approximation of F with the input difference $\alpha \in \mathcal{G}$ and the output mask $\chi \in \hat{\mathcal{H}}$, we have*

$$DL_{\alpha, \chi}^F = \sum_{\rho \in \hat{\mathcal{K}}} DL_{\alpha, \rho}^{F_1} \|\mathcal{C}_{\rho, \chi}^{F_2}\|^2,$$

as shown in Fig. 6.

Proof. Applying Lemma 21, we have

$$DL_{\alpha, \chi}^F = \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \|\mathcal{C}_{\psi, \chi}^F\|^2.$$

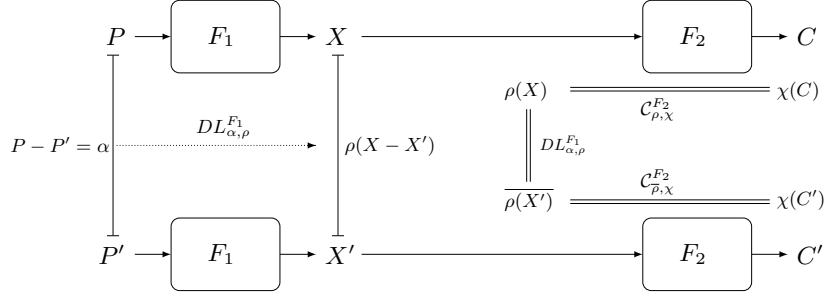


Fig. 6: A high-level illustration of a differential-linear approximation.

Due to the differential independence, according to Lemma 22, we get

$$\begin{aligned}
& \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \|\mathcal{C}_{\bar{\psi}, \chi}^F\|^2 \\
&= \sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \sum_{\rho \in \hat{\mathcal{K}}} \|\mathcal{C}_{\bar{\psi}, \rho}^{F_1}\|^2 \|\mathcal{C}_{\rho, \chi}^{F_2}\|^2 \\
&= \sum_{\rho \in \hat{\mathcal{K}}} \left(\sum_{\psi \in \hat{\mathcal{G}}} \bar{\psi}(\alpha) \|\mathcal{C}_{\bar{\psi}, \rho}^{F_1}\|^2 \right) \|\mathcal{C}_{\rho, \chi}^{F_2}\|^2 \\
&= \sum_{\rho \in \hat{\mathcal{K}}} DL_{\alpha, \rho}^{F_1} \|\mathcal{C}_{\rho, \chi}^{F_2}\|^2. \quad (\text{Lemma 21})
\end{aligned}$$

Then, we can generalize Theorem 6 (Equation (5) in [BDKW19]) as the following.

Corollary 2. *Let $(\mathcal{G}, +)$, $(\mathcal{K}, +)$, $(\mathcal{L}, +)$ and $(\mathcal{H}, +)$ be finite Abelian groups, and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function such that $F = F_2 \circ F_m \circ F_1$, where $F_1 : \mathcal{G} \rightarrow \mathcal{L}$, $F_m : \mathcal{L} \rightarrow \mathcal{K}$ and $F_2 : \mathcal{K} \rightarrow \mathcal{H}$ are differentially independent. Then, for the correlation of the differential-linear approximation of F with the input difference $\alpha \in \mathcal{G}$ and the output mask $\chi \in \hat{\mathcal{H}}$, we have:*

$$DL_{\alpha, \chi}^F = \sum_{\delta \in \mathcal{L}, \rho \in \hat{\mathcal{K}}} \Pr[\alpha \xrightarrow{F_1} \delta] DL_{\delta, \rho}^{F_m} \|\mathcal{C}_{\rho, \chi}^{F_2}\|^2.$$

Proof. Firstly, we have

$$\begin{aligned}
DL_{\alpha, \chi}^F &= \sum_{\beta \in \mathcal{H}} \chi(\beta) \Pr[\alpha \xrightarrow{F} \beta] = \sum_{\delta \in \mathcal{L}} \sum_{\beta \in \mathcal{H}} \chi(\beta) \Pr[\alpha \xrightarrow{F_1} \delta \xrightarrow{F_2 \circ F_m} \beta] \\
&= \sum_{\delta \in \mathcal{L}} \sum_{\beta \in \mathcal{H}} \chi(\delta) \Pr[\alpha \xrightarrow{F_1} \delta] \Pr[\delta \xrightarrow{F_2 \circ F_m} \beta] \\
&= \sum_{\delta \in \mathcal{L}} \Pr[\alpha \xrightarrow{F_1} \delta] \sum_{\beta \in \mathcal{H}} \chi(\beta) \Pr[\delta \xrightarrow{F_2 \circ F_m} \beta] = \sum_{\delta \in \mathcal{L}} \Pr[\alpha \xrightarrow{F_1} \delta] DL_{\delta, \chi}^{F_2 \circ F_m}.
\end{aligned}$$

Applying Theorem 8, we have

$$DL_{\alpha,\chi}^F = \sum_{\delta \in \mathcal{L}} \Pr[\alpha \xrightarrow{F_1} \delta] DL_{\delta,\chi}^{F_2 \circ F_m} = \sum_{\delta \in \mathcal{L}} \sum_{\rho \in \hat{\mathcal{K}}} \Pr[\alpha \xrightarrow{F_1} \delta] DL_{\delta,\rho}^{F_m} \|C_{\rho,\chi}^{F_2}\|^2.$$

For the differential-linear approximation of $\mathbb{F}_p^n$, we can derive the closed formula, which is powerful evidence that the definition of differential-linear approximation of $\mathbb{F}_p^n$ in [XCWW23] is a natural extension from $\mathbb{F}_2^n$ to $\mathbb{F}_p^n$. This is because the derived formula given in Corollary 3 over $\mathbb{F}_p^n$ reduces to the form given in Theorem 6 (the binary field case) when $p = 2$.

Corollary 3. *For the cipher $E : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$, where $E = E_1 \circ E_m \circ E_0$, if E_0 , E_m and E_1 are independent, the correlation of a differential-linear distinguisher over $\mathbb{F}_p$ with the input difference α and the output linear mask γ can be computed as*

$$DL_{\alpha,\lambda}^E = \sum_{\delta, \gamma \in \mathbb{F}_q^n} \Pr[\alpha \xrightarrow{E_0} \delta] DL_{\delta,\gamma}^{E_m} \|C_{\gamma,\lambda}^{E_1}\|^2, \quad (16)$$

In addition, based on Theorem 8, we have the following results over finite Abelian groups, which is the generalization of Theorem 3 and Corollary 2 in [BLN17].

Corollary 4. *For $F = F_2 \circ F_1$, suppose that $F_1 : \mathcal{G} \rightarrow \mathcal{K}$ is bijective, there is one strong linear approximation over the function $F_2 : K \rightarrow H$ with masks $(\rho, \chi) \in (\hat{\mathcal{K}}, \hat{\mathcal{H}})$, and that the correlations of all other approximations are of equal absolute value. Then, when $|\mathcal{K}|$ is large enough, according to Theorem 8, we have*

$$DL_{\alpha,\chi}^F \approx DL_{\alpha,\rho}^{F_1} \|C_{\rho,\chi}^{F_2}\|^2. \quad (17)$$

Moreover, for some $\delta \in \mathcal{K}$, if then the differential probabilities $\Pr[\alpha \xrightarrow{F_1} \delta']$ for $\delta' \neq \delta$ are equally small and the input mask $\rho \in \hat{\mathcal{K}}$ of the strong linear approximation over the function F_2 satisfies $\rho \neq 1$, then we have

$$DL_{\alpha,\chi}^F \approx \rho(k) \Pr[\alpha \xrightarrow{F_1} \delta] \|C_{\rho,\chi}^{F_2}\|^2.$$

Proof. For $\rho' \neq \rho$, according to Theorem 2, we have

$$\|C_{\rho',\chi}^{F_2}\|^2 = \frac{1 - \|C_{\rho,\chi}^{F_2}\|^2}{|K| - 1}.$$

On the other hand, for $\alpha \neq 0$, we have

$$\begin{aligned} \sum_{\rho \in \hat{\mathcal{K}}} DL_{\alpha,\rho}^{F_1} &= \sum_{\rho \in \hat{\mathcal{K}}} \frac{1}{|\mathcal{G}|} \sum_{\beta \in \mathcal{K}} \rho(\beta) \delta_{F_1}(\alpha, \beta) \\ &= \frac{1}{|\mathcal{G}|} \sum_{\beta \in \mathcal{K}} \delta_{F_1}(\alpha, \beta) \sum_{\rho \in \hat{\mathcal{K}}} \rho(\beta) \\ &= \frac{|\mathcal{K}|}{|\mathcal{G}|} \delta_{F_1}(\alpha, 0) = 0. \end{aligned}$$

Thus, according to Theorem 8, we have

$$\begin{aligned} DL_{\alpha,\chi}^F &= DL_{\alpha,\rho}^{F_1} \|\mathcal{C}_{\rho,\chi}^{F_2}\|^2 + \frac{-(1 - \|\mathcal{C}_{\rho,\chi}^{F_2}\|^2) DL_{\alpha,\rho}^{F_1}}{|\mathcal{K}| - 1} \\ &= DL_{\alpha,\rho}^{F_1} \|\mathcal{C}_{\rho,\chi}^{F_2}\|^2 \left(1 + \frac{1}{|\mathcal{K}| - 1}\right) - \frac{DL_{\alpha,\rho}^{F_1}}{|\mathcal{K}| - 1}. \end{aligned}$$

When $|\mathcal{K}|$ is large enough, we have

$$0 < \frac{\|DL_{\alpha,\rho}^{F_1}\|}{|\mathcal{K}| - 1} < \frac{1}{|\mathcal{K}| - 1} \approx 0.$$

Thus,

$$DL_{\alpha,\chi}^F \approx DL_{\alpha,\rho}^{F_1} \|\mathcal{C}_{\rho,\chi}^{F_2}\|^2.$$

Moreover, for some $\beta \in \mathcal{K}$, if the differential probabilities $\Pr[\alpha \xrightarrow{F_1} \beta']$ for $\beta' \neq \beta$ are equally small, then

$$\Pr[\alpha \xrightarrow{F_1} \beta'] = \frac{1 - \Pr[\alpha \xrightarrow{F_1} \beta]}{|\mathcal{K}| - 2}.$$

Then, for $\rho \neq \hat{1}$, we have

$$\begin{aligned} DL_{\alpha,\rho}^{F_1} &= \frac{1}{|G|} \sum_{\delta \in \mathcal{K}} \rho(\delta) \delta_{F_1}(\alpha, \delta) \\ &= (\Pr[\alpha \xrightarrow{F_1} \beta] - \frac{1 - \Pr[\alpha \xrightarrow{F_1} \beta]}{|\mathcal{K}| - 2}) \rho(\beta) + \frac{1 - \Pr[\alpha \xrightarrow{F_1} \beta]}{|\mathcal{K}| - 2} \sum_{\delta \in \mathcal{K}} \rho(\delta) \\ &= \Pr[\alpha \xrightarrow{F_1} \beta] \rho(\beta) - \frac{1 - \Pr[\alpha \xrightarrow{F_1} \beta]}{|\mathcal{K}| - 2} \rho(\beta). \end{aligned}$$

Thus, when $|\mathcal{K}|$ is large enough, we have

$$\frac{1 - \Pr[\alpha \xrightarrow{F_1} \beta]}{|\mathcal{K}| - 2} \rho(\beta) \approx 0.$$

In this case, we thereby have

$$DL_{\alpha,\chi}^F \approx \rho(\beta) \Pr[\alpha \xrightarrow{F_1} \beta] \|\mathcal{C}_{\rho,\chi}^{F_2}\|^2,$$

which completes the proof.

6 Summary and Future Work

Based on the new links we established between linear cryptanalysis and differential cryptanalysis over general finite Abelian groups, we are able to give an exact

expression of the correlation of differential-linear approximations over arbitrary finite Abelian groups $\mathbb{F}_p^n$ under the differential independence assumption. This extends a series of works including Blondeau, Leander and Nyberg [BLN17], Bar-On, Dunkelman, Keller, and Weizman [BDKW19], Xu, Chen, Wang, and Wei [XCWW23], etc. On the other hand, we gave general results of linear approximations over arbitrary finite Abelian groups, which thereby generalized the work of Pott [Pot04], and the Sidelnikov-Chabaud-Vaudenay bound [CV94] from $\mathbb{F}_2^n$ to finite Abelian groups. In summary, we give the lower bound of λ_F for all different cases (for the definition of $\rho_{\mathcal{G}, \mathcal{H}}$, see Definition 8):

1. When $|\mathcal{H}|$ divide $|\mathcal{G}|$ and $\rho_{\mathcal{G}, \mathcal{H}} \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$,

$$\lambda_F \geq |\mathcal{G}|^{-\frac{1}{2}}.$$

2. When $|\mathcal{H}| \nmid |\mathcal{G}|$ and $\rho_{\mathcal{G}, \mathcal{H}} \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$,

$$\lambda_F \geq \left(\frac{|\mathcal{H}|(|\mathcal{G}| - 1)}{|\mathcal{G}|^3(|\mathcal{H}| - 1)} \left(|\mathcal{H}| \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor^2 + 2r \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + r \right) + \frac{|\mathcal{H}| - |\mathcal{G}|}{|\mathcal{G}|(|\mathcal{H}| - 1)} \right)^{\frac{1}{2}},$$

where $|\mathcal{G}| = |\mathcal{H}| \cdot \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor + r$, $0 < r < |\mathcal{H}|$.

3. When $\rho_{\mathcal{G}, \mathcal{H}} > \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ and $\rho_{\mathcal{G}, \mathcal{H}}$ divide $|\mathcal{G}|$,

$$\lambda_F \geq \left(\frac{|\mathcal{H}|(|\mathcal{G}| - 1)}{|\mathcal{G}|^2(|\mathcal{H}| - 1)} \rho_{\mathcal{G}, \mathcal{H}} + \frac{|\mathcal{H}| - |\mathcal{G}|}{|\mathcal{G}|(|\mathcal{H}| - 1)} \right)^{\frac{1}{2}}.$$

4. When $\rho_{\mathcal{G}, \mathcal{H}} > \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor$ and $\rho_{\mathcal{G}, \mathcal{H}} \nmid |\mathcal{G}|$,

$$\lambda_F \geq \left(\frac{|\mathcal{H}|(|\mathcal{G}| - 1)}{|\mathcal{G}|^3(|\mathcal{H}| - 1)} ((m - h) \cdot c^2 + h \cdot (c + 1)^2) + \frac{|\mathcal{H}| - |\mathcal{G}|}{|\mathcal{G}|(|\mathcal{H}| - 1)} \right)^{\frac{1}{2}},$$

where $|\mathcal{G}| = m \cdot \rho_{\mathcal{G}, \mathcal{H}} + n$, $0 < n < \rho_{\mathcal{G}, \mathcal{H}}$ and $|\mathcal{G}| = c \cdot m + h$, $0 < h < m$.

It is worth exploring more applications of the new results and methods in this paper to practical ciphers, e.g., Poseidon [GKR⁺21], Rubato [HKL⁺22], Elisabeth [CHMS22], MiMC [AGR⁺16], GMiMC [AGP⁺19], HadesMiMC [GLR⁺20]. Furthermore, for some specific finite Abelian groups $\mathcal{G}$ and $\mathcal{H}$, can we construct a series of functions meeting the lower bound of λ_F we derived?

Beyne et al. [BV24, BV25] recently revealed a deeper connection between "linear approximations" over finite monoids and integral attacks. Will there be a similar lower bound for functions $F : M \rightarrow N$ between finite monoids M and N ? This is left as another open problem.

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A Some properties about λ_F

Theorem 9. *Let $(\mathcal{G}, +)$ and $(\mathcal{H}, +)$ be two finite Abelian groups and $F : \mathcal{G} \rightarrow \mathcal{H}$ be a function. $\lambda_F = |\mathcal{G}|^{-\frac{1}{2}}$ if and only if $\delta_F(\alpha, \beta) = \frac{|\mathcal{G}|}{|\mathcal{H}|}$ for all $\beta \in \mathcal{H}$, $\alpha \in \mathcal{G}$ and $\alpha \neq 0$, where $\rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}} \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor, |\mathcal{H}| \mid |\mathcal{G}|$.*

Proof. $\Leftarrow$ Combing with Lemma 12, Lemma 13 Lemma 19, and Lemma 20, the theorem holds.

$\Rightarrow$ If $\lambda_F = |\mathcal{G}|^{-\frac{1}{2}}$, then we can know that $\|\mathcal{C}_{\psi, \chi}^F\| = |\mathcal{G}|^{-\frac{1}{2}}$ for all $\chi \in \hat{\mathcal{H}}$, $\chi \neq \mathbf{1}$ and $\psi \in \hat{\mathcal{G}}$ according to Theorem 2. Then, we can see that

$$\sum_{\psi \in \hat{\mathcal{G}}} \sum_{\chi \in \hat{\mathcal{H}}, \chi \neq \mathbf{1}_{\mathcal{H}}} \|\mathcal{C}_{\psi, \chi}^F\|^4 = \frac{|\mathcal{H}| - 1}{|\mathcal{G}|}. \quad (18)$$

At this time, according to Theorem 1, we have

$$\sum_{\alpha \in \mathcal{G}, \alpha \neq 0} \sum_{\beta \in \mathcal{H}} \delta_F(\alpha, \beta)^2 = \frac{|\mathcal{G}|^2(|\mathcal{G}| - 1)}{|\mathcal{H}|}. \quad (19)$$

Due to Equation (6), for any $\alpha \in \mathcal{G}$, $\alpha \neq 0$, we have

$$\sum_{\beta \in \mathcal{H}} \delta_F(\alpha, \beta)^2 \geq \xi_{\mathcal{G}, \mathcal{H}}.$$

And from Theorem 3, it's easy to verify $\xi_{\mathcal{G}, \mathcal{H}} \geq \frac{|\mathcal{G}|^2}{|\mathcal{H}|}$.

Due to Equation (19), for any $\alpha \in \mathcal{G}$, $\alpha \neq 0$, we have

$$\sum_{\beta \in \mathcal{H}} \delta_F(\alpha, \beta)^2 = \frac{|\mathcal{G}|^2}{|\mathcal{H}|}.$$

Thus, according to Lemma 12 and Definition 7, we have $\delta_F(\alpha, \beta) = \frac{|\mathcal{G}|}{|\mathcal{H}|}$ for all $\beta \in \mathcal{H}$, $\alpha \in \mathcal{G}$ and $\alpha \neq 0$, where $\rho_{\mathcal{G}, \mathcal{H}}^{\mathcal{F}} \leq \left\lfloor \frac{|\mathcal{G}|}{|\mathcal{H}|} \right\rfloor, |\mathcal{H}| \mid |\mathcal{G}|$.